Care Commons EHR

ONC Health IT Certification Readiness
Software Manual

CY2026 Base EHR Definition

The Node on FHIR Project

https://github.com/node-on-fhir/core

July 14, 2026

Certification Status

This manual documents **certification readiness**, not an issued certification. Care Commons EHR has *not* been certified under the ONC Health IT Certification Program, and no certification is claimed or implied. This document exists to initiate that process: it records the system's self-assessment against the CY2026 Base EHR criteria (45 CFR 170.315), backed by a behavioral test suite, real test-run screenshots, reproducible build coordinates, and a lockfile-derived SBOM in the chapters that follow.

The (g)(10) Standardized API surface has been exercised against the ONC-approved Inferno test suite with passing results; those results will be re-run and formally recorded as part of an ACB submission. Formal certification requires testing and issuance by an ONC-Authorized Certification Body (ONC-ACB); engaging one is the next step this document supports.

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Chapter 0

Installation, Runtime Environment, and Configuration

This chapter records the exact environment, dependencies, and configuration used to build the system and generate the certification test results in this manual. Version numbers are drawn from the repository as of the commit cited in the Reproducibility section; treat that commit plus `package-lock.json` as the authoritative build inputs.

0.1 System Requirements

Component	Requirement
Operating system	macOS (Apple Silicon or Intel) for development; Linux (Debian/Ubuntu) for CI and production; Windows via WSL2
Meteor	3.4 (`.meteor/release` = `METEOR@3.4`)
Node.js	22.22.0 — bundled by Meteor 3.4 as the app runtime; a system Node $\geq$ 20 (CI uses 20.11.0) for npm / CLI tooling
MongoDB	6.0 or newer — Meteor's bundled development Mongo is 7.0.16; CI runs the `mongo:6.0` image
Browser (E2E)	Google Chrome / Chrome for Testing, with a version-matched ChromeDriver (the runs in this manual used 149.x)
Browser (app)	Any modern evergreen browser (Chrome, Edge, Firefox, Safari)
Desktop (opt.)	Electron via `npmPackages/desktop-lattice` — not required for certification testing

Table 1: System requirements

0.2 Package Inventory

Major dependencies, grouped by purpose. Versions are the declared ranges in `package.json`; exact resolved versions are pinned in `package-lock.json`.

Purpose	Package	Version
FHIR server	(in-repo) `server/FhirEndpoints.js`, SearchParametersEngine	—
	`@node-on-fhir/us-core` (US Core 7.0 profiles)	7.0.0
	`fhirpath`	4.8.0
SMART / Auth	`@node-oauth/express-oauth-server`	4.1.5
	`jsonwebtoken`	9.0.2
	`pkijs` + `asn1js` (X.509 / Backend Services)	3.2.4 / 3.0.5
	`bcrypt`	6.0.0
Data / Mongo	Meteor Mongo (server + Minimongo)	(Meteor 3.4)
	`simpl-schema`, `ajv` (FHIR JSON Schema)	— / 8.20.0
	`lodash`, `moment`	4.18.1 / 2.29.4
Testing	`nightwatch` (E2E)	3.12.1
	`chromedriver`	146.0.6

Purpose	Package	Version
	`meteortesting:mocha` (unit)	(Meteor)
UI	`react` / `react-dom`	18.2.0
	`@mui/material` / `@mui/icons-material`	5.16.7
C-CDA / QRDA	`@node-on-fhir/ccda-export` (C-CDA gen + receive)	(in-repo)
	`@node-on-fhir/quality-measures` (QRDA I export)	(in-repo)
	`fqm-execution` (FHIR measure engine)	1.8.5
	`xml2js` (XML parse / build)	0.6.2
Terminology	CodeSystems / ValueSets collections + US Core profiles	—
	code systems: CDCREC, LOINC, SNOMED CT, RxNorm (by reference)	—
DICOM	`@cornerstonejs/core`	1.78.0
	`@cornerstonejs/dicom-image-loader` / `tools`	1.78.0
Build tooling	Meteor 3.4 bundler	3.4
	`@rspack/core` (Rspack)	1.7.1
	`tectonic` (LaTeX — this manual only)	0.16.x

0.3 Installation

Meteor is the prerequisite; it provisions the bundled Node and MongoDB.

```
# 1. Install Meteor 3.x (once, system-wide)
curl https://install.meteor.com/ | sh

# 2. Clone and install
git clone https://github.com/node-on-fhir/core.git
cd core
npm install # installs npm workspaces + symlinks workflow packages

# 3. Run (starts the app + bundled MongoDB on http://localhost:3000)
npm start # equivalent to: meteor run
```

There is no separate `npm run dev` script; `npm start` maps to `meteor run`. A first run compiles the bundle and may take several minutes.

0.4 Environment Variables

The following variables configure the runtime; use safe placeholders in shared environments and never commit real secrets. Development / test values:

```
NODE_ENV=development # hard gate for dev auto-login
DEV_AUTO_LOGIN=true # enable dev auto-login
DEV_AUTO_USERNAME=demouser # dev account username
DEV_AUTO_PASSWORD=<dev-password> # dev account password (server-side only)
DEV_AUTO_PATIENT_ID=<fhir-id> # optional: bind dev user to a Patient
TEST_RUN=1 # gates destructive test-only operations
ENABLE_AUTOPUBLISH=true # enable autopublish.* publications
INITIALIZE_SEARCH_PARAMETERS=true # compile the FHIR SearchParametersEngine
```

```
                                 # (REQUIRED for (g)(7)/(g)(9) search)
EXTRA_WORKFLOWS=@node-on-fhir/... # comma-separated workflow packages to load
MONGO_URL=mongodb://localhost:27017/meteor # override bundled dev Mongo
                                        # (CI: mongodb://mongo:27017/meteor_test)
CHROMEDRIVER_PATH=/path/to/chromedriver # match your local Chrome version
```

Server configuration that is not an environment variable is supplied via a Meteor settings JSON file (`-settings`). Public keys land in `Meteor.settings.public` (client-visible); private keys stay server-side. Relevant fields (safe placeholders):

```
{
  "public": {
    "title": "Care Commons",
    "fhirVersion": "R4",
    "devAutoLoginEnabled": true,
    "modules": {
      "patientDemographics": { "raceEthnicity": true }
    }
  },
  "private": {
    "fhir": {
      "rest": {
        "Patient": { "interactions": ["read","create","update","delete","search"],
                    "search": true, "publication": true }
      },
      "disableOauth": true
    },
    "accessControl": {
      "enableBasicAuth": false,
      "systemSecret": "<CHANGE-ME-do-not-commit>"
    }
  }
}
```

Security note. The certification/TDD settings file ships with `disableOauth:  true` so unauthenticated REST calls are granted a `noauth` role — convenient for a self-contained test run, but never a production posture. Production deployments must enable OAuth / SMART, set a real `systemSecret` (or migrate to JWT/JWK Backend Services), and keep settings files containing secrets out of version control.

0.5 Canonical Certification Configuration

The certification test run in this manual was launched with the following exact command — environment variables, the workflow-package set, and the settings file. This is the reproducible entry point for the whole suite:

```
TEST_RUN=1 \
ENABLE_AUTOPUBLISH=true \
DEV_AUTO_LOGIN=true \
DEV_AUTO_USERNAME=demouser \
DEV_AUTO_PASSWORD=<dev-password> \
INITIALIZE_SEARCH_PARAMETERS=true \
```

```
EXTRA_WORKFLOWS=@node-on-fhir/order-catalog,@node-on-fhir/implantable-devices,@node-on-
    fhir/ccda-export,@node-on-fhir/prescription-benefit,@node-on-fhir/decision-support,
    @node-on-fhir/quality-measures,@node-on-fhir/secure-messaging \
  meteor run --settings settings/settings.honeycomb.tdd.json
```

The settings file `settings/settings.honeycomb.tdd.json` enables dev auto-login, the per-resource FHIR REST interactions (`private.fhir.rest.*`), and the race/ethnicity demographics gate (`public.modules.patientDemographics.raceEthnicity = true`, required for the (a)(5) test). Meteor reads `-settings` into `Meteor.settings` at boot. The seven `EXTRA_WORKFLOWS` packages supply the CPOE catalog, implantable-device registry, C-CDA export/receive, real-time prescription benefit, decision support, quality measures, and secure messaging surfaces exercised by the criteria chapters.

0.6 Test Execution

Run the full Base EHR suite against a running server with the aggregate runner:

```
./certification/tdd/base_ehr/run-base-ehr-tests.sh
```

Or run a single criterion directly (the form used to capture the per-chapter results and screenshots in this manual):

```
CHROMEDRIVER_PATH=/path/to/chromedriver \
  npx nightwatch --config nightwatch.circle.conf.js \
  certification/tdd/base_ehr/170.315.<x>.<y>.test.js
```

Each behavioral test drives real record / change / access behaviors against FHIR resources per the ONC test procedures; a criterion is **Verified** when its test passes green and a **Gap** when a test deliberately fails on a documented missing capability. See the Certification Testing Notes appendix for the full suite status and the relationship to Inferno.

0.7 Reproducibility

The results, screenshots, and status table in this manual were generated with:

Input	**Value**
Repository	`github.com/node-on-fhir/core` (branch `baseehr-testcoverage`)
Commit SHA	`c4ad354131cea9801a585b76ff04a629939eab24` (documentation-only changes may follow this commit)
`package-lock.json`	lockfileVersion 3; SHA-256 `a1a97949761c95fcde77784cc2a4c647ed9748443744c31965f7d3fd4a73782b` (4,121 locked packages, excluding the root project entry)
Node.js	22.22.0 (Meteor 3.4 runtime); 20.11.0 (CI tooling)
MongoDB	7.0.16 (bundled dev); 6.0 (CI)
Meteor	3.4
Browser	Chrome for Testing with version-matched ChromeDriver (149.x)

Table 3: Reproducibility inputs

To reproduce, check out the commit above, restore dependencies from `package-lock.json` with `npm ci`, launch with the Canonical Certification Configuration, and run the test-execution commands above.

Part I

Base EHR Certification

Overview

This Part documents the Base EHR certification criteria per the ONC Health IT Certification Program, updated for the **CY2026 Base EHR definition** (45 CFR 170.102, as revised by the HTI final rules). The CPOE family (a)(1)–(a)(3) requires at least one member; the remaining rows are each required (some with future compliance dates, noted below).

Certification Criteria Summary

The **Status** column reflects the outcome of the behavioral Nightwatch test suite under `certification/tdd/base_e` (one test file per criterion). **Verified** = the capability is built and its behavioral tests pass green. **Gap** = a behavioral test exists and deliberately fails on a documented missing capability (the suite is a live certification punch-list — see the per-criterion chapters and the Certification Testing Notes appendix). **Inferno** = validated with the ONC-approved Inferno tool (previously run to green; to be re-run and formally recorded for ACB submission), not by this suite.

Scope of the BDD feature files: the embedded Gherkin features are the *complete* behavioral specification for each criterion, and deliberately include forward-looking scenarios beyond the current regulatory floor (for example, the pronoun and inpatient mortality-data scenarios in (a)(5)). A criterion's **Verified** status attests to the scenarios exercised by its behavioral Nightwatch test, which may be a subset of the full feature file; the feature files are the specification, the tests are the evidence.

Criterion	Title	Status
§ 170.315(a)(1)	CPOE - Medications	Verified
§ 170.315(a)(2)	CPOE - Laboratory	Verified
§ 170.315(a)(3)	CPOE - Diagnostic Imaging	Verified
§ 170.315(a)(5)	Demographics	Verified
§ 170.315(a)(14)	Implantable Device List	Verified
§ 170.315(b)(1)	Transitions of Care (C-CDA)	Verified
§ 170.315(b)(4)	Real-Time Prescription Benefit	Verified
§ 170.315(b)(11)	Decision Support Interventions	Verified
§ 170.315(c)(1)	Clinical Quality Measures - Record & Export	Verified
§ 170.315(g)(7)	Application Access - Patient Selection	Verified
§ 170.315(g)(9)	Application Access - All Data Request	Verified
§ 170.315(g)(10)	Standardized API	Inferno
§ 170.315(h)(1)	Direct Project (or (h)(2))	Gap

Table 4: CY2026 Base EHR Certification Criteria — test-verified status (11 verified, 1 documented gap, 1 external/Inferno)

The remaining documented gap is (h)(1) Direct messaging, which has no operational transport (SMTP/S-MIME/HISP) and uses simulated certificate trust; it is detailed in its chapter and tracked in the project's engineering ledger. The former (a)(5), (b)(1), and (c)(1) gaps have since been remediated: (a)(5) now records race/ethnicity per CDCREC as US Core extensions (settings-gated — see that chapter); (b)(1) now populates the C-CDA from the real patient record and adds an inbound receive/validate/display path; and (c)(1) now exports a QRDA Category I document per § 170.205(h)(2).

Note on future compliance dates: (b)(4) Real-Time Prescription Benefit is an HTI-4 criterion required by 1/1/2028, and (b)(11) Decision Support Interventions has a Privacy & Security update due 12/31/2027. Both are tested at their current baseline; absent future-dated portions are treated as informational, not gaps.

Legacy / non-core criteria documented separately: § 170.315(a)(4) Drug Interaction Checks, § 170.315(a)(9) Clinical Decision Support (**expired** 1/1/2025, replaced by (b)(11)), and § 170.315(e)(1) View, Download, Transmit are retained as reference chapters below but are not part of the CY2026 Base EHR core set above.

Chapter 1

§ 170.315(a)(1) - CPOE Medications

1.1 Criterion Information

1.1.1 Maturity Level

VERIFIED (behavioral tests green)

1.1.2 Regulatory Text

From *45 CFR § 170.315(a)(1)*:

> **(1) Computerized provider order entry—medications.**
> **(i)** Enable a user to record, change, and access medication orders.
> **(ii)** Optional. Include a "reason for order" field.

1.2 Behavior-Driven Development Specification

Listing 1.1: 170.315(a)(1) - CPOE Medications

```
# certification/bdd/170.315-a-1-cpoe-medications.feature
# §170.315(a)(1) - Computerized Provider Order Entry - Medications

# REGULATORY TEXT FROM 45 CFR §170.315(a)(1)
#
# (1) Computerized provider order entry--medications.
#
# (i) Enable a user to record, change, and access medication orders.
#
# (ii) Optional. Include a "reason for order" field.

@170.315-a-1 @cpoe @medications @clinical
Feature: Computerized Provider Order Entry - Medications
  As a provider
  I want to create and manage medication orders

  Background:
    Given I am authenticated as a provider
    And a patient record is selected
    And I am on /order-catalog

  Scenario: Record a new medication order
    Given the patient requires "Metformin␣500␣MG␣Oral␣Tablet"
    When I open the OrderCatalog and select the medications order type
    And select the medication from the catalog
    And provide an optional "reason␣for␣order" of "Type␣2␣diabetes␣mellitus"
    And submit the order
    Then the system shall create a FHIR MedicationRequest with status "active" and intent
        "order"
    And the MedicationRequest shall carry the RxNorm-coded medication, requester, and
       dosage instructions
    And the MedicationRequest shall be inserted into the MedicationRequests collection
    And an AuditEvent shall record the order creation
```

```
Scenario: Change an existing medication order
  Given the patient has a pending MedicationRequest
  When I change the MedicationRequest status to "on-hold" and priority to "urgent"
  Then the modified values shall be persisted in the MedicationRequests collection
  And the change shall be retrievable via the medication orders API

Scenario: Access existing medication orders
  Given the patient has one or more MedicationRequests
  When I navigate to /medication-requests
  Then the system shall update the MedicationRequests subscription by patientId
  And the MedicationRequestsTable shall display the patient's medication orders
  And I shall be able to open a MedicationRequest to view its details
```

1.3 Screenshots

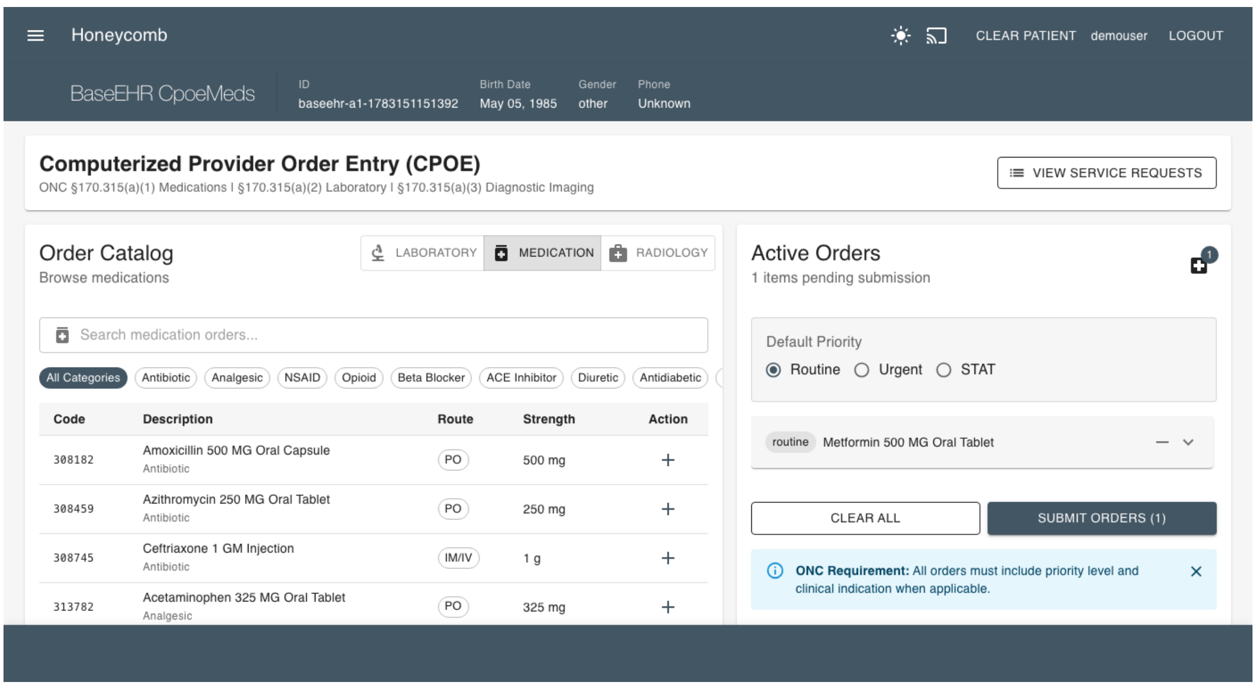

Figure 1.1: CPOE Medications - Order Entry Interface

Chapter 2

§ 170.315(a)(2) - CPOE Laboratory

2.1 Criterion Information

2.1.1 Maturity Level

VERIFIED (behavioral tests green)

2.1.2 Regulatory Text

From *45 CFR § 170.315(a)(2)*:

> **(2) Computerized provider order entry—laboratory.**
> **(i)** Enable a user to record, change, and access laboratory orders.
> **(ii)** Optional. Include a "reason for order" field.

2.2 Behavior-Driven Development Specification

Listing 2.1: 170.315(a)(2) - CPOE Laboratory

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-a-2-cpoe-
    laboratory.feature
# §170.315(a)(2) - Computerized Provider Order Entry - Laboratory

# REGULATORY TEXT FROM 45 CFR §170.315(a)(2)
#
# (2) Computerized provider order entry--laboratory.
#
# (i) Enable a user to record, change, and access laboratory orders.
#
# (ii) Optional. Include a "reason for order" field.

@170.315-a-2 @cpoe @laboratory @clinical
Feature: Computerized Provider Order Entry - Laboratory
  As a provider
  I want to create and manage laboratory orders

  Background:
    Given I am authenticated as a provider
    And a patient record is selected
    And I am on /order-catalog/laboratory

  Scenario: Create a new laboratory order
    Given the patient requires a "Complete␣Blood␣Count"
    When I open the OrderCatalog on /order-catalog
    And select a PlanDefinition for "CBC␣with␣differential"
    Then the system shall reference the PlanDefinition.id of "CBC␣with␣differential"
    And look up the associated actions and ActivityDefinitions
    And create a new resource based on ActivityDefinition.kind (typically a
        ServiceRequest)
    And the ServiceRequest shall include all necessary details for specimen collection
    And the ServiceRequest shall be inserted into the ServiceRequests collection

  Scenario: Change an existing laboratory order
```

```
    Given the patient has a pending ServiceRequest for "Basic␣Metabolic␣Panel"
    When I change the ServiceRequest to "Comprehensive␣Metabolic␣Panel"
    And the modified ServiceRequest shall be saved in the ServiceRequests collection

  Scenario: Access existing laboratory orders
    Given the patient has multiple ServiceRequests in various states
    When I request to view the patient's ServiceRequests
    Then the system shall update the ServiceRequests subscription by patientId
    And the ServiceRequestsTable shall show current status (pending, completed, cancelled
        ) for each request
    And I shall be able to view results for completed orders


Background:
    Given I am authenticated as a lab tech

  Scenario: Access workqueue
    Given no patient record is selected
    And I am on /service-requests
    Then the system shall enable access to all ServiceRequests of type "laboratory"
    And the orders shall show current status
    And I shall be able to open the ServiceRequest records
    And update their status and other fields
```

2.3 Screenshots

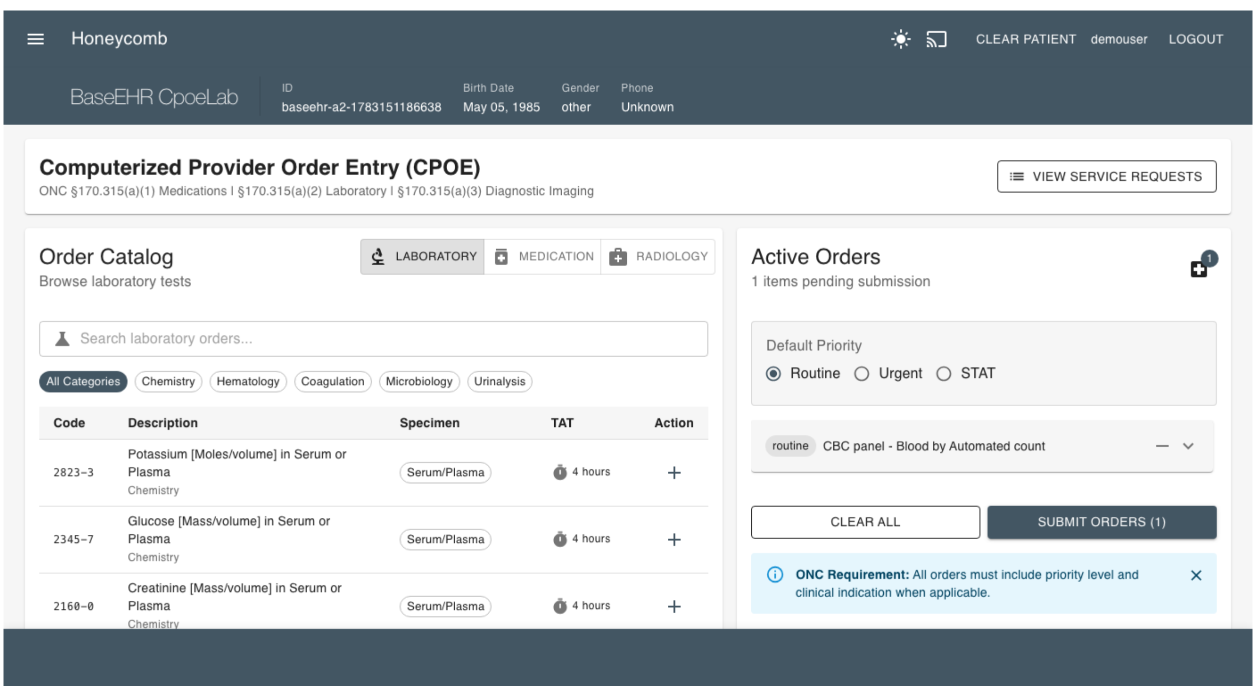

Figure 2.1: CPOE Laboratory - Order Entry Interface

Chapter 3

§ 170.315(a)(3) - CPOE Diagnostic Imaging

3.1 Criterion Information

3.1.1 Maturity Level

VERIFIED (behavioral tests green)

3.1.2 Regulatory Text

From *45 CFR § 170.315(a)(3)*:

> **(3) Computerized provider order entry—diagnostic imaging.**
> **(i)** Enable a user to record, change, and access diagnostic imaging orders.
> **(ii)** Optional. Include a "reason for order" field.

3.2 Behavior-Driven Development Specification

Listing 3.1: 170.315(a)(3) - CPOE Diagnostic Imaging

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-a-3-cpoe-
    diagnostic-imaging.feature
# §170.315(a)(3) - Computerized Provider Order Entry - Diagnostic Imaging

# REGULATORY TEXT FROM 45 CFR §170.315(a)(3)
#
# (3) Computerized provider order entry--diagnostic imaging.
#
# (i) Enable a user to record, change, and access diagnostic imaging orders.
#
# (ii) Optional. Include a "reason for order" field.

@170.315-a-3 @cpoe @diagnostic-imaging @clinical
Feature: Computerized Provider Order Entry - Diagnostic Imaging
  As an ordering provider
  I want to electronically create and manage diagnostic imaging orders
  So that I can order appropriate imaging studies efficiently

  Background:
    Given I am authenticated as a physician
    And a patient record is selected
    And I am on /order-catalog/diagnostic-imaging

  Scenario: Create a new diagnostic imaging order
    Given the patient requires a chest X-ray
    When I search the PlanDefinitions collection for "Chest␣X-ray␣PA␣and␣lateral"
    Then the system shall open a new ServiceRequest
    And the ServiceRequest shall include imaging modality and anatomical location
    And the new record will be inserted into the ServiceRequests collection

  Scenario: Change an existing diagnostic imaging order
    Given the patient has a pending ServiceRequest for "Chest␣X-ray␣PA␣and␣lateral"
    When I open the ServiceRequest and change it to "CT␣abdomen␣and␣pelvis␣with␣contrast"
        and save
```

```
    Then the system shall update the record in the ServiceRequests collection
    And it will be viewable at /service-requests

  Scenario: Optionally include reason for diagnostic imaging order
    Given I am creating a new ServiceRequest for "MRI␣lumbar␣spine"
    Then the system may include a "reason␣for␣order" field

  Scenario: Access existing diagnostic imaging
    Given a patient record is selected
    And I am on /imaging-studies
    And the patient has multiple imaging orders and completed studies (CT abdomen pelvis;
         MRI lumbar)
    Then the system shall enable access to all ImagingStudies for the selected patient
    And the orders shall show current status
    And I shall be able to access imaging reports and images (DocumentReferences)

  Background:
    Given I am authenticated as a rad tech

  Scenario: Access workqueue
    Given no patient record is selected
    And I am on /service-requests
    Then the system shall enable access to all ServiceRequests of type "imaging"
    And the orders shall show current status
    And I shall be able to open the ServiceRequest records
    And update their status and other fields

  Scenario: Access existing diagnostic imaging
    Given a patient record is selected
    And I am on /imaging-studies
    And the patient has multiple imaging orders and completed studies (CT abdomen pelvis;
         MRI lumbar)
    Then the system shall enable access to all ImagingStudies for the selected patient
    And the orders shall show current status
    And I shall be able to access imaging reports and images (DocumentReferences)
```

3.3 Screenshots

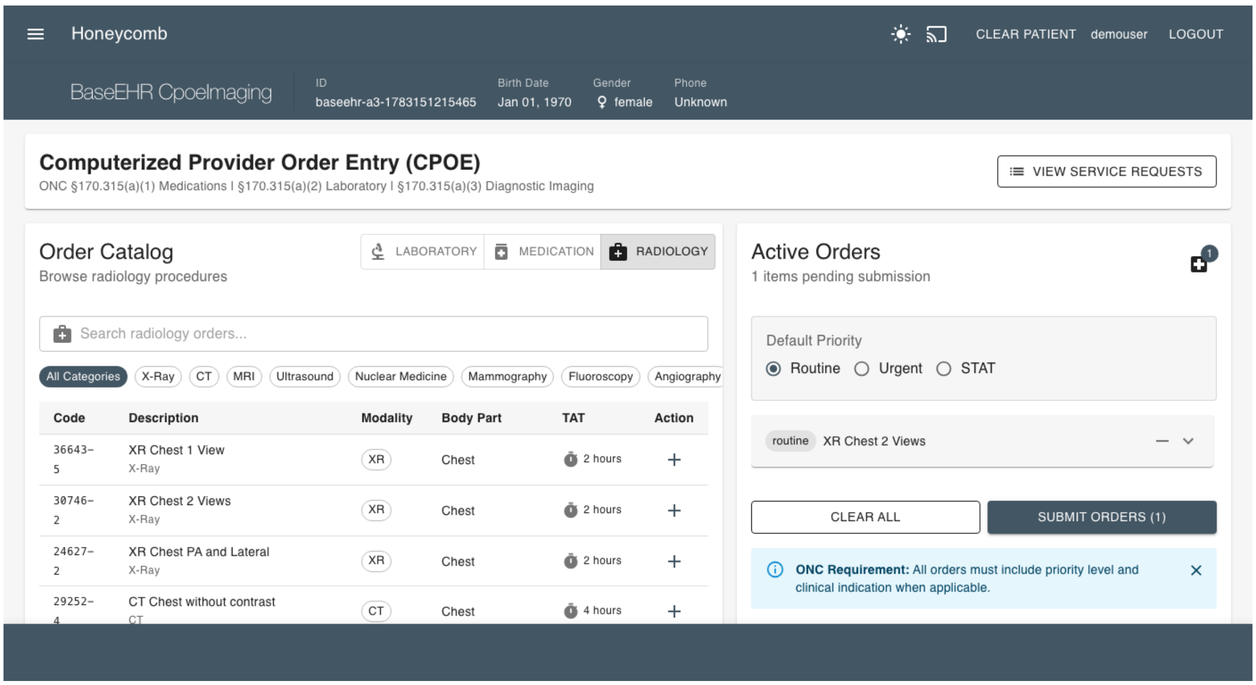

Figure 3.1: CPOE Diagnostic Imaging - Order Entry Interface

Chapter 4

§ 170.315(a)(4) - Drug Interaction Checks

4.1 Criterion Information

4.1.1 Maturity Level

IMPLEMENTED

4.1.2 Regulatory Text

From *45 CFR § 170.315(a)(4)*:

> **(4) Drug-drug, drug-allergy interaction checks for CPOE —**
> **(i) Interventions.** Before a medication order is completed and acted upon during computerized provider order entry (CPOE), interventions must automatically indicate to a user drug-drug and drug-allergy contraindications based on a patient's medication list and medication allergy list.
> **(ii) Adjustments.**
> **(A)** Enable the severity level of interventions provided for drug-drug interaction checks to be adjusted.
> **(B)** Limit the ability to adjust severity levels in at least one of these two ways:
> **(1)** To a specific set of identified users.
> **(2)** As a system administrative function.

4.2 Behavior-Driven Development Specification

Listing 4.1: 170.315(a)(4) - Drug Interaction Checks

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-a-4-drug-
    interactions.feature
# §170.315(a)(4) - Drug-Drug, Drug-Allergy Interaction Checks for CPOE

# REGULATORY TEXT FROM 45 CFR §170.315(a)(4)
#
# (4) Drug-drug, drug-allergy interaction checks for CPOE --
#
# (i) Interventions. Before a medication order is completed and acted upon during
    computerized provider order entry (CPOE), interventions must automatically indicate
    to a user drug-drug and drug-allergy contraindications based on a patient's
    medication list and medication allergy list.
#
# (ii) Adjustments.
#
# (A) Enable the severity level of interventions provided for drug-drug interaction
    checks to be adjusted.
#
# (B) Limit the ability to adjust severity levels in at least one of these two ways:
#
# (1) To a specific set of identified users.
#
# (2) As a system administrative function.

@170.315-a-4 @cpoe @drug-interactions @safety @clinical
```

```
Feature: Drug-Drug and Drug-Allergy Interaction Checks for CPOE
  As a prescribing provider
  I want automatic alerts for potential drug interactions
  So that I can prevent adverse drug events and improve patient safety

  Background:
    Given I am authenticated as a provider
    And a patient record is selected
    And the patient has an active medication list
    And the patient has a medication allergy list
    And I am on /order-catalog/medications


  Scenario: Automatic drug-drug interaction checking during CPOE
    Given the patient is currently taking "Warfarin␣5mg␣daily"
    When I am on /order-catalog/medications
    Then an RxNorm search bar will display, with RxNorm accordion table underneath with
        top-10 records
    When I select the medication request for "Aspirin␣325mg␣daily"
    Then the row will expand, with action buttons
    And the system shall automatically indicate drug-drug contraindications with an alert
    And the alert shall describe the interaction between Warfarin and Aspirin

  Scenario: Automatic drug-allergy interaction checking during CPOE
    Given the patient has a documented allergy to "Penicillin"
    When I am on /order-catalog/medications
    Then an RxNorm search bar will display, with RxNorm accordion table underneath with
        top-10 records
    When I complete the medication request for "Amoxicillin␣500mg"
    Then the row will expand, with action buttons
    Then the system shall automatically indicate drug-allergy contraindications with an
        alert
    And the alert shall identify the cross-reactivity concern

  Scenario: Display interventions based on active medication list
    Given the patient's active medication list contains multiple medications
    When I search for medications
    Then the system shall check against all medications in the active medication list
    And all drug-drug interactions shall be displayed as chips or icons in the accordion
        table

  Scenario: Display interventions based on medication allergy list
    Given the patient's allergy list contains documented allergies
    When I search for medications
    Then the system shall check against all allergies in the medication allergy list
    And all relevant drug-allergy contraindications shall be displayed as chips or icons
        in the accordion table

  Scenario: Enable severity level adjustment for drug-drug interactions
    Given I am a system administrator
    When I access the /drug-drug-interaction-settings
    Then the system shall enable adjustment of severity levels
    And severity levels shall control which interactions trigger alerts
```

```
Scenario: Display high-severity drug-drug interaction
  Given the interaction checking system is configured with severity levels
  And the patient is taking "MAO␣Inhibitor"
  When I attempt to order "SSRI␣antidepressant"
  Then the system shall display high-severity interaction alert
  And the alert shall require acknowledgment

Scenario: Display moderate-severity drug-drug interaction
  Given the interaction checking system is configured with severity levels
  And the patient is taking "Atorvastatin"
  When I attempt to order "Clarithromycin"
  Then the system shall display moderate-severity interaction alert
  And the alert shall provide clinical guidance

Scenario: Provider override of drug interaction alert
  Given a drug-drug interaction alert is displayed
  When I review the alert and determine the benefit outweighs risk
  Then the system shall allow me to 'override' the alert
  And the override shall expand a collapsed clinical reasoning input
  And the override shall be recorded in the audit log

Scenario: Multiple simultaneous interaction alerts
  Given the patient has complex medication regimen
  When I select a medication that interacts with multiple existing medications or
      allergies
  Then the system shall display all relevant interactions in the expansion panel
  And interactions shall be prioritized by severity
```

Chapter 5

§ 170.315(a)(5) - Demographics

5.1 Criterion Information

5.1.1 Maturity Level

VERIFIED (behavioral tests green)

5.1.2 Regulatory Text

From *45 CFR § 170.315(a)(5)*:

> **(5) Patient demographics and observations.**
> **(i)** Enable a user to record, change, and access patient demographic and observations data including race, ethnicity, preferred language, sex, sex parameter for clinical use, sexual orientation, gender identity, name to use, pronouns, and date of birth.

(Full regulatory text abbreviated for brevity. See BDD specification for complete requirements.)

5.1.3 Implementation

The behavioral test records, changes, and accesses name, date of birth, sex (US Core birth-sex, with a decline-to-specify option), and preferred language (RFC 5646), and now **race and ethnicity** per the CDCREC code system aggregated to OMB categories, with a declined-to-specify option. The patient form provides multi-select Race and Ethnicity fields, and the patient-write methods preserve the complex (nested sub-extension) US Core `us-core-race` / `us-core-ethnicity` extensions.

Settings gate (international deployments): collecting race/ethnicity is legally forbidden in some jurisdictions, so it is opt-in — the fields render, and the extensions persist, only when `settings.public.modules.patientDemographics.raceEthnicity` is enabled; when disabled the fields are hidden and the server strips the extensions defensively. US certification builds enable the flag.

Regulatory note (CCG v1.3, updated 05/2026): consistent with Executive Order 14168 and OPM guidance, sexual orientation, gender identity, sex parameter for clinical use, name to use, and pronouns are *not* required for (a)(5); the sex requirement reduces to recording Female or Male (plus decline).

5.2 Behavior-Driven Development Specification

Listing 5.1: 170.315(a)(5) - Demographics

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-a-5-
    demographics-observations.feature
# §170.315(a)(5) - Patient Demographics and Observations

# REGULATORY TEXT FROM 45 CFR §170.315(a)(5)
#
# (5) Patient demographics and observations.
#
# (i) Enable a user to record, change, and access patient demographic and observations
    data including race, ethnicity, preferred language, sex, sex parameter for clinical
    use, sexual orientation, gender identity, name to use, pronouns, and date of birth.
#
```

```
# (A) Race and ethnicity.
#
# (1) Enable each one of a patient's races to be recorded in accordance with, at a
    minimum, the standard specified in §170.207(f)(3) and whether a patient declines to
    specify race.
#
# (2) Enable each one of a patient's ethnicities to be recorded in accordance with, at a
    minimum, the standard specified in §170.207(f)(3) and whether a patient declines to
    specify ethnicity.
#
# (3) Aggregate each one of the patient's races and ethnicities recorded in accordance
    with paragraphs (a)(5)(i)(A)(1) and (2) of this section to the categories in the
    standard specified in §170.207(f)(1).
#
# (B) Preferred language. Enable preferred language to be recorded in accordance with the
     standard specified in §170.207(g)(2) and whether a patient declines to specify a
    preferred language.
#
# (C) Sex. Enable sex to be recorded in accordance with the standard specified in §
    170.207(n)(1) for the period up to and including December 31, 2025; or §170.207(n)(2)
    .
#
# (D) Sexual orientation. Enable sexual orientation to be recorded in accordance with, at
     a minimum, the version of the standard specified in §170.207(o)(1) for the period up
     to and including December 31, 2025; or §170.207(o)(3), as well as whether a patient
    declines to specify sexual orientation.
#
# (E) Gender identity. Enable gender identity to be recorded in accordance with, at a
    minimum, the version of the standard specified in §170.207(o)(2) for the period up to
     and including December 31, 2025; or §170.207(o)(3), as well as whether a patient
    declines to specify gender identity.
#
# (F) Sex Parameter for Clinical Use. Enable at least one sex parameter for clinical use
    to be recorded in accordance with, at a minimum, the version of the standard
    specified in §170.207(n)(3). Conformance with this paragraph is required by January
    1, 2026.
#
# (G) Name to Use. Enable at least one preferred name to use to be recorded. Conformance
    with this paragraph is required by January 1, 2026.
#
# (H) Pronouns. Enable at least one pronoun to be recorded in accordance with, at a
    minimum, the version of the standard specified in §170.207(o)(4). Conformance with
    this paragraph is required by January 1, 2026.
#
# (ii) Inpatient setting only. Enable a user to record, change, and access the
    preliminary cause of death and date of death in the event of mortality.

@170.315-a-5 @demographics @patient-data @clinical
Feature: Patient Demographics and Observations
  As a healthcare provider
  I want to record comprehensive patient demographic and observation data
  So that I can provide culturally competent and personalized care

  Background:
```

```gherkin
  Given I am authenticated as a provider
  And a patient record is selected

# ------ Race and Ethnicity ------

Scenario: Record patient's race using standard vocabulary
  Given I am recording patient demographics
  When I record the patient's race
  Then the system shall enable recording per §170.207(f)(3) standard
  And the system shall support granular race categories
  And the system shall enable recording multiple races

Scenario: Record multiple races for a patient
  Given a patient identifies with multiple racial categories
  When I record the patient's races
  Then the system shall enable each one of the patient's races to be recorded
  And each race shall be stored as discrete data element
  And all recorded races shall be retrievable

Scenario: Record patient decline to specify race
  Given I am recording patient demographics
  When the patient declines to specify their race
  Then the system shall enable recording that patient declines to specify race
  And this shall be stored as distinct from missing data

Scenario: Record patient's ethnicity using standard vocabulary
  Given I am recording patient demographics
  When I record the patient's ethnicity
  Then the system shall enable recording per §170.207(f)(3) standard
  And the system shall support Hispanic/Latino ethnicity categories

Scenario: Record multiple ethnicities for a patient
  Given a patient identifies with multiple ethnic categories
  When I record the patient's ethnicities
  Then the system shall enable each one of the patient's ethnicities to be recorded
  And each ethnicity shall be stored as discrete data element

Scenario: Record patient decline to specify ethnicity
  Given I am recording patient demographics
  When the patient declines to specify their ethnicity
  Then the system shall enable recording that patient declines to specify ethnicity
  And this shall be stored as distinct from missing data

Scenario: Aggregate race and ethnicity to standard categories
  Given granular race and ethnicity data has been recorded
  When aggregating for reporting purposes
  Then the system shall aggregate to categories per §170.207(f)(1)
  And aggregation shall maintain data integrity
  And original granular data shall be preserved

# ------ Preferred Language ------

Scenario: Record patient's preferred language
  Given I am recording patient demographics
```

```
    When I record the patient's preferred language
    Then the system shall enable recording per §170.207(g)(2) standard
    And the system shall use ISO 639-2 language codes

  Scenario: Record patient decline to specify preferred language
    Given I am recording patient demographics
    When the patient declines to specify preferred language
    Then the system shall enable recording that patient declines to specify
    And this shall be stored as distinct from missing data

  # ------ Sex ------

  Scenario: Record patient's sex (until December 31, 2025)
    Given the current date is before December 31, 2025
    When I record the patient's sex
    Then the system shall enable recording per §170.207(n)(1) standard
    And the system shall support administrative sex categories

  Scenario: Record patient's sex (after December 31, 2025)
    Given the current date is after December 31, 2025
    When I record the patient's sex
    Then the system shall enable recording per §170.207(n)(2) standard
    And the system shall support updated sex categories

  # ------ Sex Parameter for Clinical Use (Required by January 1, 2026) ------

  Scenario: Record sex parameter for clinical use
    Given the current date is on or after January 1, 2026
    When I record clinical observations requiring sex parameter
    Then the system shall enable recording at least one sex parameter for clinical use
    And the recording shall be per §170.207(n)(3) standard
    And the parameter shall be context-specific for clinical use

  Scenario: Support multiple sex parameters for clinical use
    Given the current date is on or after January 1, 2026
    And different clinical contexts require different parameters
    When I record sex parameters for various clinical purposes
    Then the system shall enable recording multiple sex parameters
    And each parameter shall be associated with its clinical context

  # ------ Sexual Orientation ------

  Scenario: Record patient's sexual orientation (until December 31, 2025)
    Given the current date is before December 31, 2025
    When I record the patient's sexual orientation
    Then the system shall enable recording per §170.207(o)(1) standard
    And the system shall use LOINC and SNOMED CT codes

  Scenario: Record patient's sexual orientation (after December 31, 2025)
    Given the current date is after December 31, 2025
    When I record the patient's sexual orientation
    Then the system shall enable recording per §170.207(o)(3) standard
    And the system shall support updated vocabulary
```

```gherkin
Scenario: Record patient decline to specify sexual orientation
  Given I am recording patient demographics
  When the patient declines to specify sexual orientation
  Then the system shall enable recording that patient declines to specify
  And this shall be stored as distinct from missing data

# ------ Gender Identity ------

Scenario: Record patient's gender identity (until December 31, 2025)
  Given the current date is before December 31, 2025
  When I record the patient's gender identity
  Then the system shall enable recording per §170.207(o)(2) standard
  And the system shall use standard gender identity vocabulary

Scenario: Record patient's gender identity (after December 31, 2025)
  Given the current date is after December 31, 2025
  When I record the patient's gender identity
  Then the system shall enable recording per §170.207(o)(3) standard
  And the system shall support updated vocabulary

Scenario: Record patient decline to specify gender identity
  Given I am recording patient demographics
  When the patient declines to specify gender identity
  Then the system shall enable recording that patient declines to specify
  And this shall be stored as distinct from missing data

# ------ Name to Use (Required by January 1, 2026) ------

Scenario: Record patient's preferred name to use
  Given the current date is on or after January 1, 2026
  When I record the patient's preferred name
  Then the system shall enable recording at least one preferred name to use
  And the preferred name shall be distinct from legal name
  And the preferred name shall be usable in clinical workflows

Scenario: Display patient's preferred name in user interface
  Given the current date is on or after January 1, 2026
  And the patient has a preferred name recorded
  When displaying patient information
  Then the system should prominently display the preferred name
  And the legal name shall remain accessible when needed

# ------ Pronouns (Required by January 1, 2026) ------

Scenario: Record patient's pronouns
  Given the current date is on or after January 1, 2026
  When I record the patient's pronouns
  Then the system shall enable recording at least one pronoun
  And the recording shall be per §170.207(o)(4) standard
  And the pronouns shall use standard vocabulary

Scenario: Support multiple pronoun sets
  Given the current date is on or after January 1, 2026
  And a patient uses different pronouns in different contexts
```

```
  When I record the patient's pronouns
  Then the system shall enable recording multiple pronoun sets
  And each pronoun set shall be stored and retrievable

# ------ Date of Birth ------

Scenario: Record patient's complete date of birth
  Given I am recording patient demographics
  When I record the patient's date of birth
  Then the system shall enable recording year, month, and day
  And the date shall be stored in standard format
  And the date shall be used for age calculations

Scenario: Handle unknown date of birth
  Given the patient's exact date of birth is unknown
  When I record demographic information
  Then the system shall enable recording null value for unknown date of birth
  And the system shall distinguish unknown from missing data

# ------ Change and Access Demographics ------

Scenario: Change patient demographic data
  Given patient demographic data has been recorded
  When demographic information changes or needs correction
  Then the system shall enable changing all demographic data elements
  And changes shall be audited with timestamp and user
  And history of changes shall be maintained

Scenario: Access patient demographic data
  Given patient demographic data has been recorded
  When I need to view patient demographics
  Then the system shall enable access to all demographic data elements
  And demographic data shall be displayed clearly
  And demographic data shall be available throughout clinical workflows

# ------ Inpatient-Only: Mortality Data ------

@inpatient-only
Scenario: Record preliminary cause of death in inpatient setting
  Given I am authenticated in an inpatient setting
  And a patient mortality has occurred
  When I record mortality information
  Then the system shall enable recording preliminary cause of death
  And the cause shall be coded appropriately
  And the information shall be available for death certificate

@inpatient-only
Scenario: Record date of death in inpatient setting
  Given I am authenticated in an inpatient setting
  And a patient mortality has occurred
  When I record mortality information
  Then the system shall enable recording date of death
  And the date shall include date and time
  And the date shall be stored in standard format
```

```
@inpatient-only
Scenario: Change mortality data in inpatient setting
  Given mortality data has been recorded
  When corrections are needed to mortality information
  Then the system shall enable changing preliminary cause of death
  And the system shall enable changing date of death
  And all changes shall be audited

@inpatient-only
Scenario: Access mortality data in inpatient setting
  Given mortality data has been recorded
  When generating death certificate or reports
  Then the system shall enable access to preliminary cause of death
  And the system shall enable access to date of death
  And the data shall be formatted appropriately for reporting
```

5.3 Screenshots

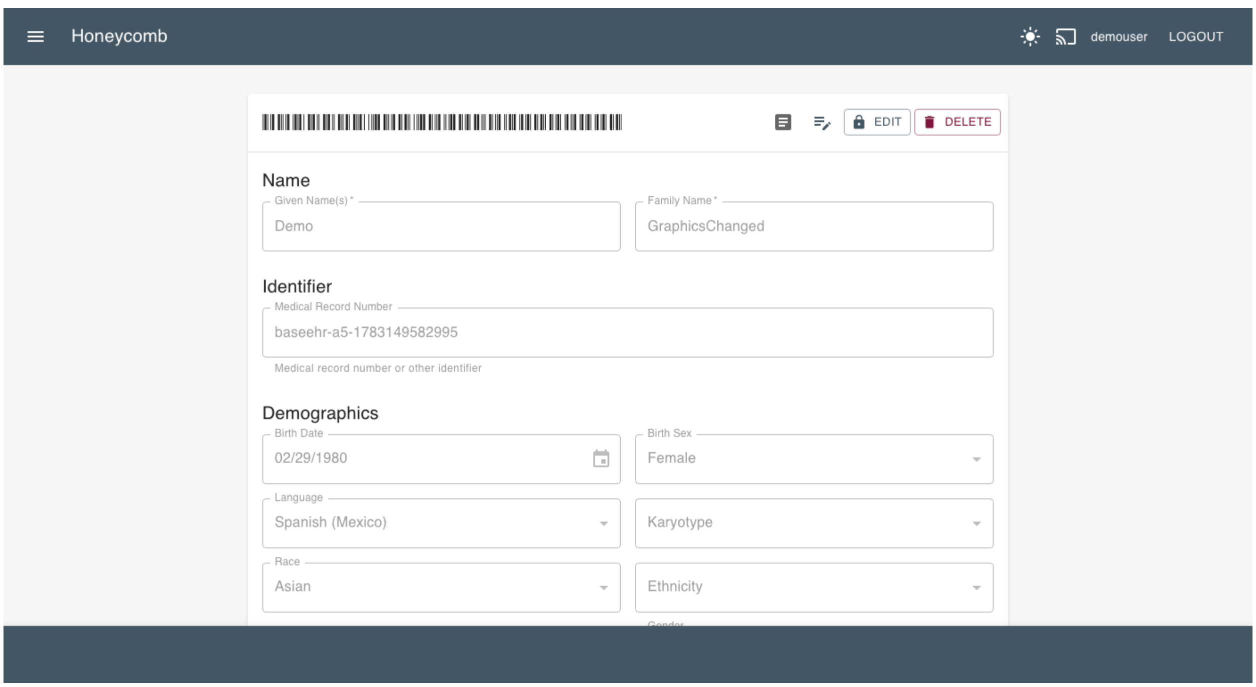

Figure 5.1: Patient Demographics - Data Entry Interface

Chapter 6

§ 170.315(a)(9) - Clinical Decision Support (EXPIRED)

6.1 Criterion Information

6.1.1 Maturity Level

IMPLEMENTED (EXPIRED)

6.1.2 Expiration Notice

CRITICAL: This criterion EXPIRED on January 1, 2025 per 45 CFR § 170.315(a)(9)(vi).

This criterion has been replaced by § 170.315(b)(11) - Decision Support Interventions (see Chapter 9).

Due to government website unavailability as of January 2025, it is unclear whether Base EHR certifications issued before the expiration date can still use this criterion (grandfathered), or if all certifications must now use § 170.315(b)(11).

6.1.3 Regulatory Text

From *45 CFR § 170.315(a)(9)*:

> **(9) Clinical decision support (CDS) —**
>
> **(i) CDS intervention interaction.** Interventions provided to a user must occur when a user is interacting with technology.
>
> (Full regulatory text abbreviated. See BDD specification for complete requirements.)
>
> **(vi) Expiration of criterion.** The adoption of this criterion for purposes of the ONC Health IT Certification Program expires on January 1, 2025.

6.2 Behavior-Driven Development Specification

Listing 6.1: 170.315(a)(9) - Clinical Decision Support

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-a-9-
    clinical-decision-support.feature
# §170.315(a)(9) - Clinical Decision Support (CDS)
# NOTE: This criterion expires January 1, 2025

# REGULATORY TEXT FROM 45 CFR §170.315(a)(9)
#
# (9) Clinical decision support (CDS) --
#
# (i) CDS intervention interaction. Interventions provided to a user must occur when a
    user is interacting with technology.
#
# (ii) CDS configuration.
#
# (A) Enable interventions and reference resources specified in paragraphs (a)(9)(iii)
    and (iv) of this section to be configured by a limited set of identified users (e.g.,
     system administrator) based on a user's role.
#
# (B) Enable interventions:
#
# (1) Based on the following data:
```

```
#
# (i) Problem list;
#
# (ii) Medication list;
#
# (iii) Allergy and intolerance list;
#
# (iv) At least one demographic specified in paragraph (a)(5)(i) of this section;
#
# (v) Laboratory tests; and
#
# (vi) Vital signs.
#
# (2) When a patient's medications, allergies and intolerance, and problems are
    incorporated from a transition of care/referral summary received and pursuant to
    paragraph (b)(2)(iii)(D) of this section.
#
# (iii) Evidence-based decision support interventions. Enable a limited set of identified
     users to select (i.e., activate) electronic CDS interventions (in addition to drug-
    drug and drug-allergy contraindication checking) based on each one and at least one
    combination of the data referenced in paragraphs (a)(9)(ii)(B)(1)(i) through (vi) of
    this section.
#
# (iv) Linked referential CDS.
#
# (A) Identify for a user diagnostic and therapeutic reference information in accordance
    at least one of the following standards and implementation specifications:
#
# (1) The standard and implementation specifications specified in §170.204(b)(3).
#
# (2) The standard and implementation specifications specified in §170.204(b)(4).
#
# (B) For paragraph (a)(9)(iv)(A) of this section, technology must be able to identify
    for a user diagnostic or therapeutic reference information based on each one and at
    least one combination of the data referenced in paragraphs (a)(9)(ii)(B)(1)(i), (ii),
     and (iv) of this section.
#
# (v) Source attributes. Enable a user to review the attributes as indicated for all CDS
    resources:
#
# (A) For evidence-based decision support interventions under paragraph (a)(9)(iii) of
    this section:
#
# (1) Bibliographic citation of the intervention (clinical research/guideline);
#
# (2) Developer of the intervention (translation from clinical research/guideline);
#
# (3) Funding source of the intervention development technical implementation; and
#
# (4) Release and, if applicable, revision date(s) of the intervention or reference
    source.
#
# (B) For linked referential CDS in paragraph (a)(9)(iv) of this section and drug-drug,
    drug-allergy interaction checks in paragraph (a)(4) of this section, the developer of
```

```
     the intervention, and where clinically indicated, the bibliographic citation of the
    intervention (clinical research/guideline).
#
# (vi) Expiration of criterion. The adoption of this criterion for purposes of the ONC
    Health IT Certification Program expires on January 1, 2025.

@170.315-a-9 @cds @clinical-decision-support @expires-2025 @clinical
Feature: Clinical Decision Support (CDS)
  As a healthcare provider
  I want evidence-based clinical decision support
  So that I can make informed treatment decisions

  Background:
    Given I am authenticated as a provider
    And a patient record is selected

  # ------ CDS Intervention Interaction ------

  Scenario: CDS interventions occur during user interaction
    Given I am actively interacting with the health IT system
    When clinical decision support is triggered
    Then interventions shall be provided during my interaction with technology
    And interventions shall not be delayed until after interaction ends

  # ------ CDS Configuration ------

  Scenario: Configure CDS interventions based on user role
    Given I am a system administrator or authorized configurator
    When I configure clinical decision support interventions
    Then the system shall enable configuration by limited set of identified users
    And configuration shall be based on user role
    And configuration capabilities shall be restricted appropriately

  Scenario: Configure CDS reference resources based on user role
    Given I am a system administrator or authorized configurator
    When I configure reference resources for CDS
    Then the system shall enable configuration by limited set of identified users
    And configuration shall be based on user role

  # ------ CDS Based on Patient Data ------

  Scenario: Enable interventions based on problem list
    Given the patient has documented problems
    When CDS evaluates patient data
    Then the system shall enable interventions based on problem list data

  Scenario: Enable interventions based on medication list
    Given the patient has active medications
    When CDS evaluates patient data
    Then the system shall enable interventions based on medication list data

  Scenario: Enable interventions based on allergy and intolerance list
    Given the patient has documented allergies and intolerances
    When CDS evaluates patient data
```

```
  Then the system shall enable interventions based on allergy and intolerance list

Scenario: Enable interventions based on demographics
  Given the patient has demographic data recorded
  When CDS evaluates patient data
  Then the system shall enable interventions based on at least one demographic
  And demographic data may include age, sex, race, ethnicity, etc.

Scenario: Enable interventions based on laboratory tests
  Given the patient has laboratory test results
  When CDS evaluates patient data
  Then the system shall enable interventions based on laboratory test data

Scenario: Enable interventions based on vital signs
  Given the patient has vital signs recorded
  When CDS evaluates patient data
  Then the system shall enable interventions based on vital signs data

Scenario: Enable interventions from incorporated transition of care data
  Given medications, allergies, and problems are incorporated from transition of care
      summary
  When the data is incorporated per §170.315(b)(2)(iii)(D)
  Then the system shall enable interventions based on incorporated data
  And interventions shall trigger automatically upon incorporation

# ------ Evidence-Based Decision Support Interventions ------

Scenario: Select evidence-based CDS interventions
  Given I am an authorized user selecting CDS interventions
  When I activate electronic CDS interventions
  Then the system shall enable selection based on each individual data element
  And the system shall enable selection based on combinations of data elements
  And data elements include problem list, medication list, allergies, demographics, lab
       tests, and vital signs

Scenario: Activate CDS intervention based on problem list
  Given I am selecting CDS interventions
  When I activate an intervention based on problem list
  Then the intervention shall utilize problem list data
  And the intervention shall provide evidence-based guidance

Scenario: Activate CDS intervention based on multiple data elements
  Given I am selecting CDS interventions
  When I activate an intervention based on combination of data elements
  Then the intervention shall utilize all specified data elements
  And the intervention shall provide integrated clinical guidance

# ------ Linked Referential CDS ------

Scenario: Identify diagnostic reference information via Infobutton
  Given I need diagnostic reference information
  When I access linked referential CDS
  Then the system shall identify information per §170.204(b)(3) standard
  And information shall be context-specific to clinical scenario
```

```gherkin
Scenario: Identify therapeutic reference information via Infobutton
  Given I need therapeutic reference information
  When I access linked referential CDS
  Then the system shall identify information per §170.204(b)(4) standard
  And information shall be context-specific to clinical scenario

Scenario: Identify reference information based on problem list
  Given the patient has documented problems
  When I access linked referential CDS
  Then the system shall identify information based on problem list data

Scenario: Identify reference information based on medication list
  Given the patient has active medications
  When I access linked referential CDS
  Then the system shall identify information based on medication list data

Scenario: Identify reference information based on demographics
  Given the patient has demographic data
  When I access linked referential CDS
  Then the system shall identify information based on demographic data

Scenario: Identify reference information based on combination of data
  Given multiple types of patient data are available
  When I access linked referential CDS
  Then the system shall identify information based on combinations of data
  And combinations may include problem list, medications, and demographics

# ------ Source Attributes for CDS ------

Scenario: Display source attributes for evidence-based interventions
  Given an evidence-based CDS intervention is active
  When I review the intervention source information
  Then the system shall enable review of bibliographic citation
  And the system shall display developer of the intervention
  And the system shall display funding source of intervention development
  And the system shall display release and revision dates

Scenario: Display source attributes for drug interaction checks
  Given drug-drug or drug-allergy interaction check is displayed
  When I review the intervention source information
  Then the system shall display developer of the intervention
  And where clinically indicated, the system shall display bibliographic citation

Scenario: Display source attributes for linked referential CDS
  Given linked referential CDS is being utilized
  When I review the reference source information
  Then the system shall display developer of the intervention
  And where clinically indicated, the system shall display bibliographic citation

Scenario: Access complete source attribute information
  Given CDS interventions are configured in the system
  When I need to review source attributes
  Then all source attributes shall be accessible
```

```
    And source attributes shall be displayed in user-friendly format
    And source attributes shall support evidence-based decision making
```

Chapter 7

§ 170.315(b)(1) - Transitions of Care

7.1 Criterion Information

7.1.1 Maturity Level

VERIFIED (behavioral tests green)

7.1.2 Regulatory Text

From *45 CFR § 170.315(b)(1)*:

> **(1) Transitions of care.**
> **(i) Send and receive via Edge Protocol.** Send and receive transition-of-care/referral summaries formatted per the Consolidated CDA (C-CDA) Release 2.1, using an Edge Protocol.
> **(ii) Validate and display.** Validate received C-CDA documents against the standard and display their contents.
> **(iii) Create.** Create a transition-of-care/referral summary as a C-CDA document, populated from the patient's record per the Common Clinical Data Set / USCDI.

(Full regulatory text abbreviated. See BDD specification for complete requirements.)

7.1.3 Implementation

The behavioral test verifies create, validate, access, and receive: a C-CDA R2.1-enveloped transition-of-care document (Transfer Summary, LOINC 18761-7) is generated with the US Realm header, CCD template identifiers, and LOINC-coded Allergies/Medications/Problems sections, **populated from the real patient record** (the document gathers Patients, AllergyIntolerances, MedicationRequests/Statements, Conditions, Observations, and Procedures for the patient); it is persisted as a FHIR `DocumentReference` with the XML as a base64 attachment plus an `AuditEvent`; and a malformed document is rejected by the validator.

An **inbound receive** path (`clinicalDocuments.receiveCCDA`) parses an externally-authored C-CDA (via XML parsing), validates the envelope, extracts patient demographics and the section list, and stores it for display in the Clinical Documents list/detail; a Receive C-CDA upload control is provided on the export page. Structured-entry *incorporation* (creating Conditions/etc. from parsed sections) and schematron-grade validation are flagged follow-ons.

7.2 Behavior-Driven Development Specification

Listing 7.1: 170.315(b)(1) - Transitions of Care

```
# certification/bdd/170.315-b-1-transitions-of-care.feature
# §170.315(b)(1) - Transitions of Care
# BDD Format using Cucumber/Gherkin Syntax

# REGULATORY TEXT FROM 45 CFR §170.315(b)(1)
#
# (1) Transitions of care --
#
# (i) Send and receive via edge protocol.
#
```

```
# (A) Send transition of care/referral summaries through a method that conforms to the
    standard specified in §170.202(d) and that leads to such summaries being processed by
     a service that has implemented the standard specified in §170.202(a)(2); and
#
# (B) Receive transition of care/referral summaries through a method that conforms to the
     standard specified in §170.202(d) from a service that has implemented the standard
    specified in §170.202(a)(2).
#
# (C) XDM processing. Receive and make available the contents of a XDM package formatted
    in accordance with the standard adopted in §170.205(p)(1) when the technology is also
     being certified using an SMTP-based edge protocol.
#
# (ii) Validate and display --
#
# (A) Validate C-CDA conformance--system performance. Demonstrate the ability to detect
    valid and invalid transition of care/referral summaries received and formatted in
    accordance with the standards specified in §170.205(a)(3), (4), and (5) for the
    Continuity of Care Document, Referral Note, and (inpatient setting only) Discharge
    Summary document templates. This includes the ability to:
#
# (1) Parse each of the document types.
#
# (2) Detect errors in corresponding "document-templates," "section-templates," and "
    entry-templates," including invalid vocabulary standards and codes not specified in
    the standards adopted in §170.205(a)(3), (4), and (5).
#
# (3) Identify valid document-templates and process the data elements required in the
    corresponding section-templates and entry-templates from the standards adopted in §
    170.205(a)(3), (4), and (5).
#
# (4) Correctly interpret empty sections and null combinations.
#
# (5) Record errors encountered and allow a user through at least one of the following
    ways to:
#
# (i) Be notified of the errors produced.
#
# (ii) Review the errors produced.
#
# (B) Display. Display in human readable format the data included in transition of care/
    referral summaries received and formatted according to the standards specified in §
    170.205(a)(3), (4), and (5).
#
# (C) Display section views. Allow for the individual display of each section (and the
    accompanying document header information) that is included in a transition of care/
    referral summary received and formatted in accordance with the standards adopted in §
    170.205(a)(3), (4), and (5) in a manner that enables the user to:
#
# (1) Directly display only the data within a particular section;
#
# (2) Set a preference for the display order of specific sections; and
#
# (3) Set the initial quantity of sections to be displayed.
#
```

```
# (iii) Create. Enable a user to create a transition of care/referral summary formatted
    in accordance with the standard specified in §170.205(a)(3), (4), and (5) using the
    Continuity of Care Document, Referral Note, and (inpatient setting only) Discharge
    Summary document templates that includes, at a minimum:
#
# (A)
#
# (1) The data classes expressed in the standards in §170.213 and in accordance with §
    170.205(a)(4), (5), and paragraphs (b)(1)(iii)(A)(3)(i) through (iii) of this section
     for the time period up to and including December 31, 2025, or
#
# (2) The data classes expressed in the standards in §170.213 and in accordance with §
    170.205(a)(4), (6), and paragraphs (b)(1)(iii)(A)(3)(i) through (iii) of this section
    , and
#
# (3) The following data classes:
#
# (i) Assessment and plan of treatment. In accordance with the "Assessment and Plan
    Section (V2)" of the standard specified in §170.205(a)(4); or in accordance with the
    "Assessment Section (V2)" and "Plan of Treatment Section (V2)" of the standard
    specified in §170.205(a)(4).
#
# (ii) Goals. In accordance with the "Goals Section" of the standard specified in §
    170.205(a)(4).
#
# (iii) Health concerns. In accordance with the "Health Concerns Section" of the standard
     specified in §170.205(a)(4).
#
# (iv) Unique device identifier(s) for a patient's implantable device(s). In accordance
    with the "Product Instance" in the "Procedure Activity Procedure Section" of the
    standard specified in §170.205(a)(4).
#
# (B) Encounter diagnoses. Formatted according to at least one of the following standards
    :
#
# (1) The standard specified in §170.207(i).
#
# (2) At a minimum, the version of the standard specified in §170.207(a)(1).
#
# (C) Cognitive status.
#
# (D) Functional status.
#
# (E) Ambulatory setting only. The reason for referral; and referring or transitioning
    provider's name and office contact information.
#
# (F) Inpatient setting only. Discharge instructions.
#
# (G) Patient matching data. First name, last name, previous name, middle name (including
     middle initial), suffix, date of birth, current address, phone number, and sex. The
    following constraints apply:
#
# (1) Date of birth constraint.
#
```

```
# (i) The year, month and day of birth must be present for a date of birth. The
    technology must include a null value when the date of birth is unknown.
#
# (ii) Optional. When the hour, minute, and second are associated with a date of birth
    the technology must demonstrate that the correct time zone offset is included.
#
# (2) Phone number constraint. Represent phone number (home, business, cell) in
    accordance with the standards adopted in §170.207(q)(1). All phone numbers must be
    included when multiple phone numbers are present.
#
# (3) Sex Constraint: Represent sex with the standard adopted in §170.207(n)(1) up to and
     including December 31, 2025; or with the standard adopted in §170.207(n)(2).

@170.315-b-1 @transitions-of-care @toc @care-coordination
Feature: Transitions of Care
  As a healthcare provider
  I want to send and receive comprehensive transition of care summaries
  So that I can ensure care continuity when patients move between care settings

  Background:
    Given I am authenticated as a provider
    And a patient record is selected

  # ------ Send Transition of Care Summary via Edge Protocol ------

  Scenario: Send transition of care summary via Direct
    Given a patient is being transitioned to another provider
    When I create and send a transition of care summary
    Then the system shall send through method conforming to §170.202(d)
    And the summary shall be processed by service implementing §170.202(a)(2)
    And the transmission shall use secure messaging

  Scenario: Receive transition of care summary via Direct
    Given another provider is sending a transition of care summary for my patient
    When the summary is transmitted
    Then the system shall receive through method conforming to §170.202(d)
    And the system shall receive from service implementing §170.202(a)(2)
    And the received summary shall be available for processing

  Scenario: Process XDM package when using SMTP-based edge protocol
    Given the technology uses SMTP-based edge protocol for Direct messaging
    When receiving an XDM package
    Then the system shall receive XDM package formatted per §170.205(p)(1)
    And the system shall make available the contents of the XDM package
    And contents shall be accessible for clinical review

  # ------ Validate C-CDA Conformance ------

  Scenario: Detect valid transition of care summary
    Given a transition of care summary is received
    When the system validates the document
    Then the system shall detect valid summaries formatted per §170.205(a)(3), (4), (5)
    And valid documents shall be processed for display and reconciliation
```

```
Scenario: Detect invalid transition of care summary
  Given a malformed transition of care summary is received
  When the system validates the document
  Then the system shall detect invalid summaries
  And errors shall be identified and reported

Scenario: Parse CCD document template
  Given a Continuity of Care Document is received
  When the system processes the document
  Then the system shall parse the CCD template
  And all sections shall be accessible

Scenario: Parse Referral Note document template
  Given a Referral Note is received
  When the system processes the document
  Then the system shall parse the Referral Note template
  And all sections shall be accessible

@inpatient-only
Scenario: Parse Discharge Summary document template
  Given I am in an inpatient setting
  And a Discharge Summary is received
  When the system processes the document
  Then the system shall parse the Discharge Summary template
  And all sections shall be accessible

Scenario: Detect errors in document-templates
  Given a transition of care summary with template errors is received
  When the system validates document structure
  Then the system shall detect errors in document-templates
  And specific template errors shall be identified

Scenario: Detect errors in section-templates
  Given a transition of care summary with section errors is received
  When the system validates document structure
  Then the system shall detect errors in section-templates
  And specific section errors shall be identified

Scenario: Detect errors in entry-templates
  Given a transition of care summary with entry errors is received
  When the system validates document structure
  Then the system shall detect errors in entry-templates
  And specific entry errors shall be identified

Scenario: Detect invalid vocabulary standards
  Given a transition of care summary with invalid codes is received
  When the system validates vocabulary
  Then the system shall detect invalid vocabulary standards
  And invalid codes shall be identified

Scenario: Detect codes not specified in standards
  Given a transition of care summary with non-standard codes is received
  When the system validates vocabulary
  Then the system shall detect codes not specified in §170.205(a)(3), (4), (5)
```

```
  And non-standard codes shall be reported

Scenario: Identify valid document-templates
  Given a properly formatted transition of care summary is received
  When the system validates the document
  Then the system shall identify valid document-templates
  And valid templates shall be processed successfully

Scenario: Process required data elements
  Given a valid transition of care summary is received
  When the system extracts data elements
  Then the system shall process data elements required in corresponding section-
      templates
  And the system shall process data elements required in corresponding entry-templates
  And all required data per §170.205(a)(3), (4), (5) shall be extracted

Scenario: Correctly interpret empty sections
  Given a transition of care summary with empty sections is received
  When the system processes the document
  Then the system shall correctly interpret empty sections
  And empty sections shall not cause processing errors

Scenario: Correctly interpret null combinations
  Given a transition of care summary with null values is received
  When the system processes the document
  Then the system shall correctly interpret null combinations
  And null values shall be handled appropriately

Scenario: Record errors encountered during validation
  Given a transition of care summary with multiple errors is received
  When the system validates the document
  Then the system shall record errors encountered
  And errors shall be stored for review

Scenario: Notify user of validation errors
  Given validation errors have been recorded
  When errors are detected
  Then the system shall enable user notification of errors produced
  And notifications shall be timely and clear

Scenario: Enable user to review validation errors
  Given validation errors have been recorded
  When a user needs to review errors
  Then the system shall enable user to review errors produced
  And error details shall be accessible
  And errors shall be presented in actionable format

# ------ Display Received Transition of Care Summary ------

Scenario: Display transition of care summary in human readable format
  Given a valid transition of care summary is received
  When I view the summary
  Then the system shall display data in human readable format
  And the display shall be formatted per §170.205(a)(3), (4), (5)
```

```gherkin
    And all included data shall be viewable

  Scenario: Display CCD in human readable format
    Given a Continuity of Care Document is received
    When I view the document
    Then all data shall be displayed in human readable format
    And the display shall be clinically useful

  Scenario: Display Referral Note in human readable format
    Given a Referral Note is received
    When I view the document
    Then all data shall be displayed in human readable format
    And referral information shall be prominent

  @inpatient-only
  Scenario: Display Discharge Summary in human readable format
    Given I am in an inpatient setting
    And a Discharge Summary is received
    When I view the document
    Then all data shall be displayed in human readable format
    And discharge information shall be prominent

  # ------ Display Individual Sections ------

  Scenario: Display individual section from transition summary
    Given a transition of care summary is received
    When I want to view a specific section
    Then the system shall allow individual display of each section
    And accompanying document header information shall be displayed

  Scenario: Directly display only data within particular section
    Given a transition of care summary with multiple sections is received
    When I select a specific section to view
    Then the system shall enable direct display of only that section's data
    And other sections shall be hidden from view

  Scenario: Set preference for display order of sections
    Given I have specific workflow preferences
    When I configure section display settings
    Then the system shall enable setting preference for display order of sections
    And my preferences shall be saved and applied

  Scenario: Set initial quantity of sections displayed
    Given I want to control information density
    When I configure section display settings
    Then the system shall enable setting initial quantity of sections displayed
    And I shall be able to expand to view additional sections

  # ------ Create Transition of Care Summary ------

  Scenario: Create CCD document template
    Given I need to send a transition of care summary
    When I create a Continuity of Care Document
    Then the system shall format per §170.205(a)(3), (4), (5)
```

```
    And the CCD shall include all required USCDI data classes per §170.213

  Scenario: Create Referral Note document template
    Given I need to send a referral
    When I create a Referral Note
    Then the system shall format per §170.205(a)(3), (4), (5)
    And the Referral Note shall include all required USCDI data classes per §170.213

  @inpatient-only
  Scenario: Create Discharge Summary document template
    Given I am in an inpatient setting
    And a patient is being discharged
    When I create a Discharge Summary
    Then the system shall format per §170.205(a)(3), (4), (5)
    And the Discharge Summary shall include all required USCDI data classes per §170.213

  Scenario: Include all USCDI data classes in transition summary
    Given I am creating a transition of care summary
    When I generate the summary
    Then the system shall include all data classes per §170.213
    And data shall be formatted per §170.205(a)(4), (5) or (6) based on date
    And all mandatory data elements shall be included

  Scenario: Include assessment and plan of treatment
    Given I am creating a transition of care summary
    When I generate the summary
    Then the system shall include assessment and plan per "Assessment␣and␣Plan␣Section␣(
        V2)"
    Or the system shall include separate "Assessment␣Section␣(V2)" and "Plan␣of␣Treatment
        ␣Section␣(V2)"
    And content shall conform to §170.205(a)(4)

  Scenario: Include goals in transition summary
    Given I am creating a transition of care summary
    When I generate the summary
    Then the system shall include goals per "Goals␣Section"
    And goals shall conform to §170.205(a)(4)

  Scenario: Include health concerns in transition summary
    Given I am creating a transition of care summary
    When I generate the summary
    Then the system shall include health concerns per "Health␣Concerns␣Section"
    And health concerns shall conform to §170.205(a)(4)

  Scenario: Include unique device identifiers in transition summary
    Given the patient has implantable devices
    When I create a transition of care summary
    Then the system shall include UDIs per "Product␣Instance" in "Procedure␣Activity␣
        Procedure␣Section"
    And UDIs shall conform to §170.205(a)(4)

  Scenario: Include encounter diagnoses in transition summary
    Given I am creating a transition of care summary
    When I generate the summary
```

```
    Then the system shall include encounter diagnoses
    And diagnoses shall be formatted per §170.207(i) or §170.207(a)(1)

  Scenario: Include cognitive status in transition summary
    Given I am creating a transition of care summary
    When I generate the summary
    Then the system shall include cognitive status
    And cognitive status shall be appropriately coded

  Scenario: Include functional status in transition summary
    Given I am creating a transition of care summary
    When I generate the summary
    Then the system shall include functional status
    And functional status shall be appropriately coded

  # ------ Patient Matching Data ------

  Scenario: Include patient matching data in transition summary
    Given I am creating a transition of care summary
    When I generate the summary
    Then the system shall include first name, last name, previous name, middle name,
        suffix
    And the system shall include date of birth
    And the system shall include current address
    And the system shall include phone number
    And the system shall include sex

  Scenario: Include complete date of birth with null for unknown
    Given I am creating a transition of care summary
    When including date of birth
    Then year, month, and day must be present for date of birth
    And the system shall include null value when date of birth is unknown

  Scenario: Optionally include time zone with date of birth
    Given I am creating a transition of care summary
    When date of birth includes time components
    Then the system may include hour, minute, and second
    And if time is included, correct time zone offset must be included

  Scenario: Represent phone numbers per standard
    Given I am creating a transition of care summary
    When including phone numbers
    Then phone numbers shall be represented per §170.207(q)(1)
    And all phone numbers (home, business, cell) must be included when multiple are
        present

  Scenario: Represent sex per standard until December 31, 2025
    Given the current date is before December 31, 2025
    When including sex in transition summary
    Then sex shall be represented per §170.207(n)(1)

  Scenario: Represent sex per updated standard after December 31, 2025
    Given the current date is after December 31, 2025
    When including sex in transition summary
```

```
  Then sex shall be represented per §170.207(n)(2)

# ------ Ambulatory Setting Specific ------

@ambulatory-only
Scenario: Include reason for referral in ambulatory setting
  Given I am in an ambulatory setting
  When I create a transition of care summary
  Then the system shall include reason for referral
  And the reason shall be clinically meaningful

@ambulatory-only
Scenario: Include referring provider information in ambulatory setting
  Given I am in an ambulatory setting
  When I create a transition of care summary
  Then the system shall include referring provider's name
  And the system shall include referring provider's office contact information

# ------ Inpatient Setting Specific ------

@inpatient-only
Scenario: Include discharge instructions in inpatient setting
  Given I am in an inpatient setting
  When I create a transition of care summary
  Then the system shall include discharge instructions
  And discharge instructions shall be comprehensive
```

7.3 Screenshots

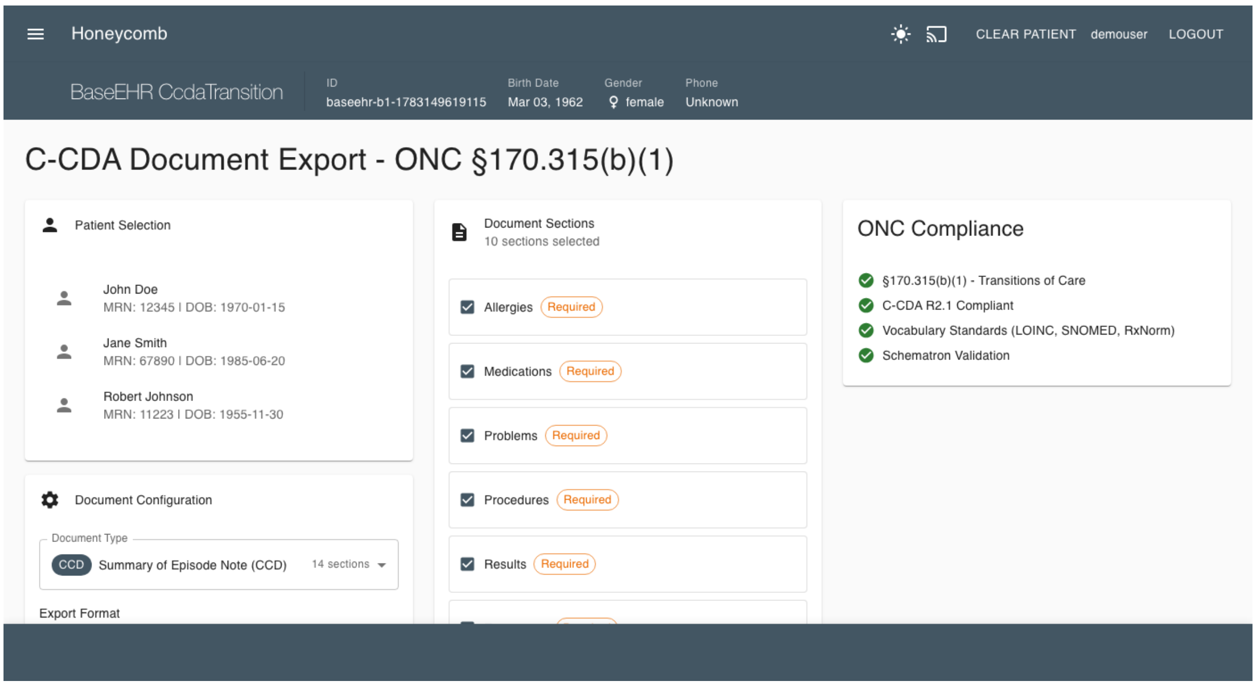

Figure 7.1: Transitions of Care - C-CDA Generation

Chapter 8

§ 170.315(b)(4) - Real-Time Prescription Benefit

8.1 Criterion Information

8.1.1 Maturity Level

VERIFIED (behavioral tests green)

8.1.2 Regulatory Text

From *45 CFR § 170.315(b)(4)*:

> **(4) Real-time prescription benefit (RTPB).**
> Enable a user, for a prescribed drug, to request and receive real-time prescription benefit information — including patient cost, formulary/coverage status, and any therapeutic alternatives — from a benefit source.

(Full regulatory text abbreviated. See BDD specification for complete requirements.)

8.1.3 Implementation Status

The behavioral test drives the full RTPB round-trip: it composes an `RTPBRequest` for a prescribed product, transacts it against a benefit responder (a formulary/PBM responder from the responder registry), and receives an `RTPBResponse` carrying the patient's cost and cheaper therapeutic alternatives (a brand statin returns generic substitutions). Both request and response XML are rendered and persisted with request/response linkage, and the completed check appears in the transaction history. A configurable live endpoint is supported alongside the built-in certification-evidence responders.

Future compliance date: (b)(4) is an HTI-4 criterion required by 1/1/2028. The current baseline (request/response, patient cost, alternatives, live-endpoint capability) is verified; the final standard-version binding is treated as informational until closer to the compliance date, not a gap.

8.2 Behavior-Driven Development Specification

A dedicated Gherkin feature file for (b)(4) is not yet authored; the executable specification is the behavioral Nightwatch test `certification/tdd/base_ehr/170.315.b.4.test.js`, which drives the request/response round-trip, patient-cost and therapeutic-alternative assertions, and the transaction-history retrieval described above.

8.3 Screenshots

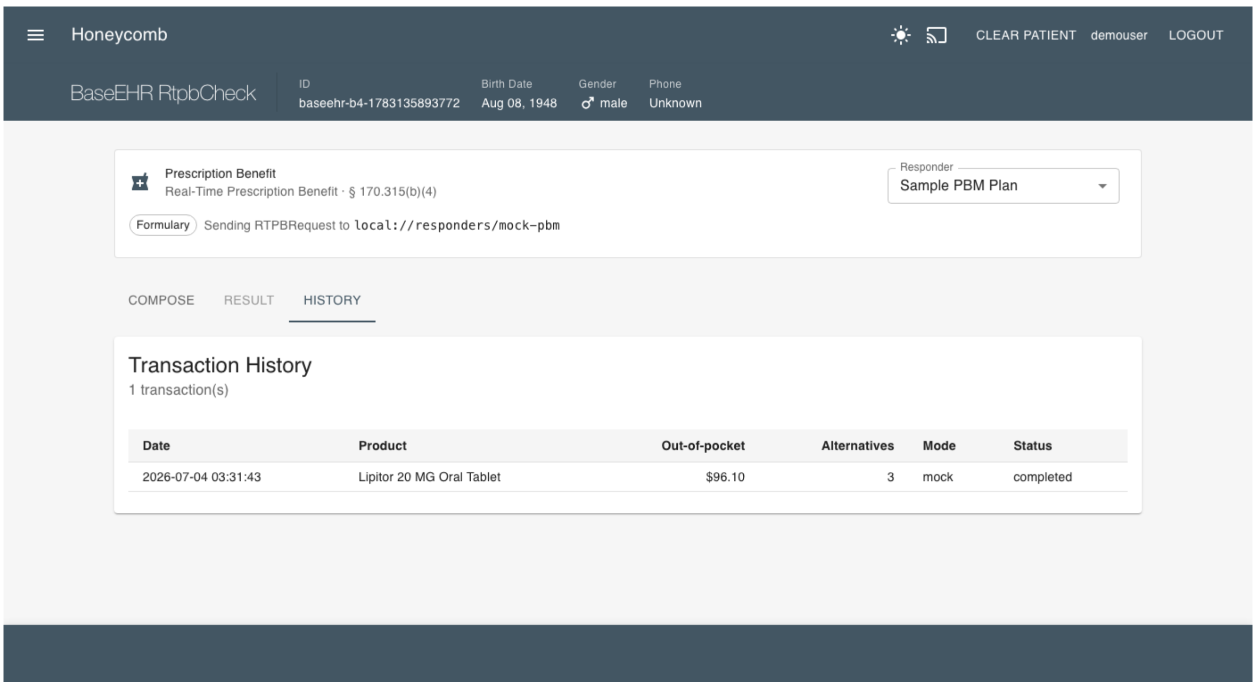

Figure 8.1: Real-Time Prescription Benefit - Cost and Alternatives

Chapter 9

§ 170.315(b)(11) - Decision Support Interventions

9.1 Criterion Information

9.1.1 Maturity Level

VERIFIED (behavioral tests green)

9.1.2 Replacement Notice

This criterion REPLACES § 170.315(a)(9) which expired January 1, 2025.

Decision Support Interventions (DSI) is an enhanced version of the legacy Clinical Decision Support criterion, adding requirements for:

- Predictive decision support interventions (AI/ML algorithms)
- Demographic data usage transparency
- Electronic intervention feedback mechanism
- Enhanced risk and fairness disclosures for predictive DSI

9.1.3 Regulatory Text

From *45 CFR § 170.315(b)(11)*:

> **(11) Decision Support Interventions.**
> (Full regulatory text includes all requirements from a-9 plus enhanced requirements for predictive interventions, demographic transparency, feedback mechanisms, and algorithm governance. See BDD specification for complete requirements.)

9.2 Behavior-Driven Development Specification

Listing 9.1: 170.315(b)(11) - Decision Support Interventions

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-b-11-
    decision-support-interventions.feature
# §170.315(b)(11) - Decision Support Interventions

# REGULATORY TEXT FROM 45 CFR §170.315(b)(11)
#
# (11) Use of date of birth as expressed in the standards in §170.213;

@170.315-b-11 @dsi @predictive-ai @algorithm-transparency @care-coordination
Feature: Decision Support Interventions
  As a healthcare provider
  I want transparent and fair decision support interventions
  So that I can make evidence-based decisions while understanding AI/algorithm influences

  Background:
    Given I am authenticated as a provider
    And I have a patient record open
    And decision support interventions are configured
```

```gherkin
# ------ Intervention Interaction ------

Scenario: Interventions occur during user interaction
  Given I am actively interacting with health IT
  When decision support is triggered
  Then interventions shall be provided during my interaction with technology
  And interventions shall occur at clinically appropriate times

# ------ Configuration ------

Scenario: Configure DSI interventions based on user role
  Given I am authorized to configure decision support
  When configuring interventions
  Then the system shall enable configuration by limited set of identified users
  And configuration shall be based on user role
  And configuration capabilities shall be appropriately restricted

Scenario: Enable interventions when data incorporated from transition of care
  Given medications, allergies, and problems are incorporated from transition summary
  When data is incorporated per §170.315(b)(2)(iii)(D)
  Then interventions shall be enabled based on incorporated data
  And interventions shall trigger appropriately

Scenario: Provide electronic feedback capability for interventions
  Given I am using evidence-based decision support interventions
  When I want to provide feedback on intervention
  Then the system shall enable me to provide electronic feedback data
  And feedback shall be collectible for selected interventions
  And limited set of users shall be able to export feedback in computable format

Scenario: Export intervention feedback data
  Given electronic feedback has been collected
  When authorized users need to analyze feedback
  Then feedback data shall be exportable in computable format
  And export shall include intervention, action taken, user feedback, user, date, and
      location
  And feedback analysis shall support intervention improvement

# ------ Evidence-Based Decision Support Intervention Selection ------

Scenario: Select evidence-based interventions using USCDI data
  Given I am authorized to select decision support interventions
  When activating evidence-based interventions
  Then interventions may use problems per §170.213
  And interventions may use medications per §170.213
  And interventions may use allergies and intolerances per §170.213
  And interventions may use demographics per §170.213
  And interventions may use laboratory data per §170.213
  And interventions may use vital signs per §170.213
  And interventions may use unique device identifiers per §170.213
  And interventions may use procedures per §170.213

# ------ Predictive Decision Support Intervention Selection ------
```

```
Scenario: Select predictive interventions using any USCDI data
  Given I am authorized to select decision support interventions
  When activating predictive decision support interventions
  Then interventions may use any data expressed in §170.213
  And predictive algorithms shall utilize appropriate data elements
  And data inputs shall be transparent

# ------ Source Attributes for Evidence-Based Interventions ------

Scenario: Display evidence-based intervention source attributes
  Given an evidence-based intervention is active
  When I review intervention source information
  Then the system shall support display of bibliographic citation
  And the system shall support display of intervention developer
  And the system shall support display of funding source for development
  And the system shall support display of release and revision dates

Scenario: Display demographic data usage in evidence-based intervention
  Given an evidence-based intervention uses demographic data
  When I review intervention details
  Then the system shall indicate use of race per §170.213
  And the system shall indicate use of ethnicity per §170.213
  And the system shall indicate use of language per §170.213
  And the system shall indicate use of sexual orientation per §170.213
  And the system shall indicate use of gender identity per §170.213
  And the system shall indicate use of sex per §170.213
  And the system shall indicate use of date of birth per §170.213

Scenario: Display SDOH and health status data usage
  Given an evidence-based intervention uses SDOH data
  When I review intervention details
  Then the system shall indicate use of SDOH data per §170.213
  And the system shall indicate use of health status assessments data per §170.213

# ------ Source Attributes for Predictive Interventions (EXTENSIVE) ------

Scenario: Display predictive intervention details and output
  Given a predictive decision support intervention is active
  When I review intervention details
  Then the system shall display name and contact information for developer
  And the system shall display funding source of technical implementation
  And the system shall display description of value produced as output
  And the system shall display whether output is prediction, classification,
      recommendation, evaluation, analysis, or other type

Scenario: Display predictive intervention purpose
  Given a predictive intervention is active
  When I review intervention purpose
  Then the system shall display intended use of intervention
  And the system shall display intended patient population(s)
  And the system shall display intended user(s)
  And the system shall display intended decision-making role (informs, augments,
      replaces clinical management)
```

```gherkin
Scenario: Display cautioned out-of-scope use
  Given a predictive intervention has usage limitations
  When I review intervention limitations
  Then the system shall display description of cautioned tasks, situations, or
      populations
  And the system shall display known risks, inappropriate settings, inappropriate uses,
       or limitations

Scenario: Display intervention development details and input features
  Given a predictive intervention is active
  When I review development details
  Then the system shall display exclusion and inclusion criteria for training data
  And the system shall display use of demographic variables as input features
  And the system shall display description of demographic representativeness in
      training data
  And the system shall display description of relevance of training data to deployed
      setting

Scenario: Display fairness process for intervention development
  Given a predictive intervention is active
  When I review fairness information
  Then the system shall display description of approach to ensure intervention output
      is fair
  And the system shall display description of approaches to manage, reduce, or
      eliminate bias

Scenario: Display external validation process
  Given a predictive intervention has been externally validated
  When I review validation information
  Then the system shall display description of data source, setting, or environment for
       validation
  And the system shall display party that conducted external testing
  And the system shall display demographic representativeness of external data
  And the system shall display description of external validation process

Scenario: Display quantitative performance measures
  Given a predictive intervention is active
  When I review performance metrics
  Then the system shall display validity in test data from training source
  And the system shall display fairness in test data from training source
  And the system shall display validity in external data
  And the system shall display fairness in external data
  And the system shall display references to evaluation of outcomes impact

Scenario: Display ongoing maintenance information
  Given a predictive intervention is deployed
  When I review maintenance information
  Then the system shall display process and frequency for monitoring validity over time
  And the system shall display validity in local data
  And the system shall display process and frequency for monitoring fairness over time
  And the system shall display fairness in local data

Scenario: Display update and validation schedule
  Given a predictive intervention is in use
```

```
  When I review update information
  Then the system shall display process and frequency for intervention updates
  And the system shall display frequency for performance correction when risks
      identified

Scenario: Indicate when optional information not available
  Given a predictive intervention source attributes
  When certain information is not available for review
  Then the system shall indicate when information not available for external validation
  And the system shall indicate when information not available for external validity
      metrics
  And the system shall indicate when information not available for external fairness
      metrics
  And the system shall indicate when information not available for outcome evaluations
  And the system shall indicate when information not available for local validity
  And the system shall indicate when information not available for local fairness
  And the system shall indicate when information not available for update schedule
      details

# ------ Access and Modify Source Attributes ------

Scenario: Access complete plain language descriptions for developer-supplied
    interventions
  Given interventions are supplied by health IT developer
  When authorized users need to review source attributes
  Then limited set of identified users shall be able to access complete descriptions
  And descriptions shall be in plain language
  And descriptions shall be up-to-date

Scenario: Modify source attributes for interventions
  Given intervention source attributes need updating
  When authorized users modify attributes
  Then limited set of identified users shall record, change, and access source
      attributes
  And modifications shall be tracked with audit trail
  And all users shall see updated information

Scenario: Record additional attributes for predictive interventions
  Given predictive interventions may have additional relevant attributes
  When documenting intervention characteristics
  Then limited set of users shall record, change, and access additional attributes
  And additional attributes supplement required attributes
  And comprehensive documentation shall be maintained

# ------ Intervention Risk Management for Predictive DSI ------

Scenario: Conduct risk analysis for predictive intervention
  Given a predictive decision support intervention is implemented
  When assessing intervention risks
  Then intervention shall be subject to analysis of potential risks and adverse impacts
  And analysis shall assess validity, reliability, robustness, fairness
  And analysis shall assess intelligibility, safety, security, and privacy

Scenario: Implement risk mitigation for predictive intervention
```

```
  Given risks have been identified for predictive intervention
  When implementing risk controls
  Then intervention shall be subject to practices to mitigate identified risks
  And mitigation shall address all significant risk categories
  And mitigation effectiveness shall be monitored

Scenario: Apply governance to predictive intervention
  Given a predictive intervention is in production use
  When managing intervention lifecycle
  Then intervention shall be subject to policies and implemented controls for
      governance
  And governance shall include how data are acquired, managed, and used
  And governance shall ensure ongoing oversight and accountability
```

9.3 Screenshots

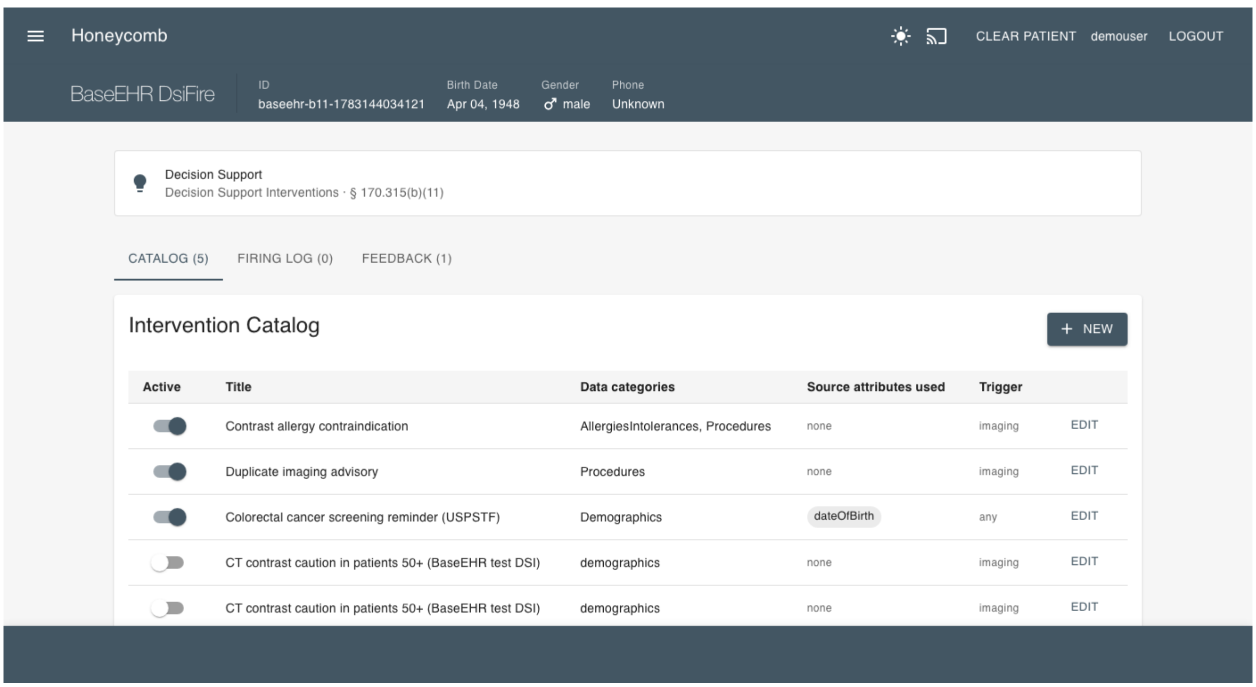

Figure 9.1: Decision Support Interventions - Configuration Interface

Chapter 10

§ 170.315(a)(14) - Implantable Device List

10.1 Criterion Information

10.1.1 Maturity Level

VERIFIED (behavioral tests green)

10.1.2 Regulatory Text

From *45 CFR § 170.315(a)(14)*:

> **(14) Implantable device list.**
> **(i)** Record Unique Device Identifiers associated with a patient's Implantable Devices.
> **(ii)** Parse the following identifiers from a Unique Device Identifier.
> (Full regulatory text abbreviated. See BDD specification for complete requirements.)

10.2 Behavior-Driven Development Specification

Listing 10.1: 170.315(a)(14) - Implantable Device List

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-a-14-
    implantable-device-list.feature
# §170.315(a)(14) - Implantable Device List

# REGULATORY TEXT FROM 45 CFR §170.315(a)(14)
#
# (14) Implantable device list.
#
# (i) Record Unique Device Identifiers associated with a patient's Implantable Devices.
#
# (ii) Parse the following identifiers from a Unique Device Identifier:
#
# (A) Device Identifier; and
#
# (B) The following identifiers that compose the Production Identifier:
#
# (1) The lot or batch within which a device was manufactured;
#
# (2) The serial number of a specific device;
#
# (3) The expiration date of a specific device;
#
# (4) The date a specific device was manufactured; and
#
# (5) For an HCT/P regulated as a device, the distinct identification code required by 21
     CFR 1271.290(c).
#
# (iii) Obtain and associate with each Unique Device Identifier:
#
# (A) A description of the implantable device referenced by at least one of the following
    :
#
```

```
# (1) The "GMDN PT Name" attribute associated with the Device Identifier in the Global
    Unique Device Identification Database.
#
# (2) The "SNOMED CTő Description" mapped to the attribute referenced in in paragraph (a)
    (14)(iii)(A)(1) of this section.
#
# (B) The following Global Unique Device Identification Database attributes:
#
# (1) "Brand Name";
#
# (2) "Version or Model";
#
# (3) "Company Name";
#
# (4) "What MRI safety information does the labeling contain?"; and
#
# (5) "Device required to be labeled as containing natural rubber latex or dry natural
    rubber (21 CFR 801.437)."
#
# (iv) Display to a user an implantable device list consisting of:
#
# (A) The active Unique Device Identifiers recorded for the patient;
#
# (B) For each active Unique Device Identifier recorded for a patient, the description of
     the implantable device specified by paragraph (a)(14)(iii)(A) of this section; and
#
# (C) A method to access all Unique Device Identifiers recorded for a patient.
#
# (v) For each Unique Device Identifier recorded for a patient, enable a user to access:
#
# (A) The Unique Device Identifier;
#
# (B) The description of the implantable device specified by paragraph (a)(14)(iii)(A) of
     this section;
#
# (C) The identifiers associated with the Unique Device Identifier, as specified by
    paragraph (a)(14)(ii) of this section; and
#
# (D) The attributes associated with the Unique Device Identifier, as specified by
    paragraph (a)(14)(iii)(B) of this section.
#
# (vi) Enable a user to change the status of a Unique Device Identifier recorded for a
    patient.

@170.315-a-14 @udi @implantable-devices @clinical
Feature: Implantable Device List
  As a healthcare provider
  I want to record and track implantable devices using Unique Device Identifiers
  So that I can ensure patient safety, support recall management, and provide accurate
     device information

  Background:
    Given I am authenticated as a provider
    And a patient record is selected
```

```
# ------ Record UDI ------

Scenario: Record Unique Device Identifier for implantable device
  Given a patient has received an implantable device
  When I record the device information
  Then the system shall enable recording of Unique Device Identifiers
  And the UDI shall be associated with the patient record
  And the UDI shall be stored in structured format

# ------ Parse UDI Components ------

Scenario: Parse Device Identifier from UDI
  Given a Unique Device Identifier has been recorded
  When the system processes the UDI
  Then the system shall parse the Device Identifier (DI)
  And the DI shall be stored as separate data element

Scenario: Parse Production Identifier components from UDI
  Given a Unique Device Identifier has been recorded
  When the system processes the UDI
  Then the system shall parse lot or batch number
  And the system shall parse serial number
  And the system shall parse expiration date
  And the system shall parse manufacture date
  And all production identifiers shall be stored as separate data elements

Scenario: Parse HCT/P distinct identification code from UDI
  Given a Unique Device Identifier for HCT/P regulated device has been recorded
  When the system processes the UDI
  Then the system shall parse distinct identification code per 21 CFR 1271.290(c)
  And the code shall be stored appropriately

# ------ Obtain Device Information from GUDID ------

Scenario: Obtain device description from GUDID
  Given a Unique Device Identifier has been entered
  When the system queries the Global Unique Device Identification Database
  Then the system shall obtain device description
  And description may be "GMDN␣PT␣Name" from GUDID
  Or description may be "SNOMED␣CT␣Description" mapped to GMDN PT Name

Scenario: Obtain device brand name from GUDID
  Given a Unique Device Identifier has been entered
  When the system queries GUDID
  Then the system shall obtain "Brand␣Name" attribute
  And the brand name shall be associated with the UDI

Scenario: Obtain device version or model from GUDID
  Given a Unique Device Identifier has been entered
  When the system queries GUDID
  Then the system shall obtain "Version␣or␣Model" attribute
  And the version/model shall be associated with the UDI
```

```
Scenario: Obtain device company name from GUDID
  Given a Unique Device Identifier has been entered
  When the system queries GUDID
  Then the system shall obtain "Company␣Name" attribute
  And the manufacturer information shall be associated with the UDI

Scenario: Obtain MRI safety information from GUDID
  Given a Unique Device Identifier has been entered
  When the system queries GUDID
  Then the system shall obtain MRI safety information
  And the information shall indicate MRI labeling content
  And MRI safety information shall be prominently available

Scenario: Obtain latex allergy information from GUDID
  Given a Unique Device Identifier has been entered
  When the system queries GUDID
  Then the system shall obtain latex labeling information per 21 CFR 801.437
  And latex information shall be prominently available for patient safety

# ------ Display Implantable Device List ------

Scenario: Display patient's implantable device list
  Given a patient has recorded Unique Device Identifiers
  When I view the implantable device list
  Then the system shall display active Unique Device Identifiers
  And the system shall display device descriptions
  And the display shall be organized and easily reviewable

Scenario: Display device description for each active UDI
  Given a patient has multiple active implantable devices
  When I view the implantable device list
  Then for each active UDI the system shall display device description
  And description shall be from GUDID per requirements
  And description shall be clinically meaningful

Scenario: Provide method to access all UDIs for patient
  Given a patient has both active and inactive device records
  When I need to view complete device history
  Then the system shall provide method to access all UDIs
  And I shall be able to distinguish active from inactive devices
  And complete device history shall be accessible

# ------ Access Individual UDI Details ------

Scenario: Access complete UDI information
  Given a patient has a recorded Unique Device Identifier
  When I select a specific UDI to review
  Then the system shall enable access to the complete UDI
  And the system shall display the device description
  And the system shall display parsed production identifiers
  And the system shall display GUDID attributes

Scenario: Access parsed identifiers for UDI
  Given a patient has a recorded Unique Device Identifier
```

```
  When I access the UDI details
  Then the system shall display lot/batch number
  And the system shall display serial number
  And the system shall display expiration date
  And the system shall display manufacture date
  And the system shall display HCT/P code if applicable

Scenario: Access GUDID attributes for UDI
  Given a patient has a recorded Unique Device Identifier
  When I access the UDI details
  Then the system shall display brand name
  And the system shall display version or model
  And the system shall display company name
  And the system shall display MRI safety information
  And the system shall display latex information

# ------ Change UDI Status ------

Scenario: Change status of implantable device
  Given a patient has a recorded Unique Device Identifier
  When the device is explanted or status changes
  Then the system shall enable changing the status of the UDI
  And status change shall be recorded with date and reason
  And device shall move from active to inactive list
```

10.3 Screenshots

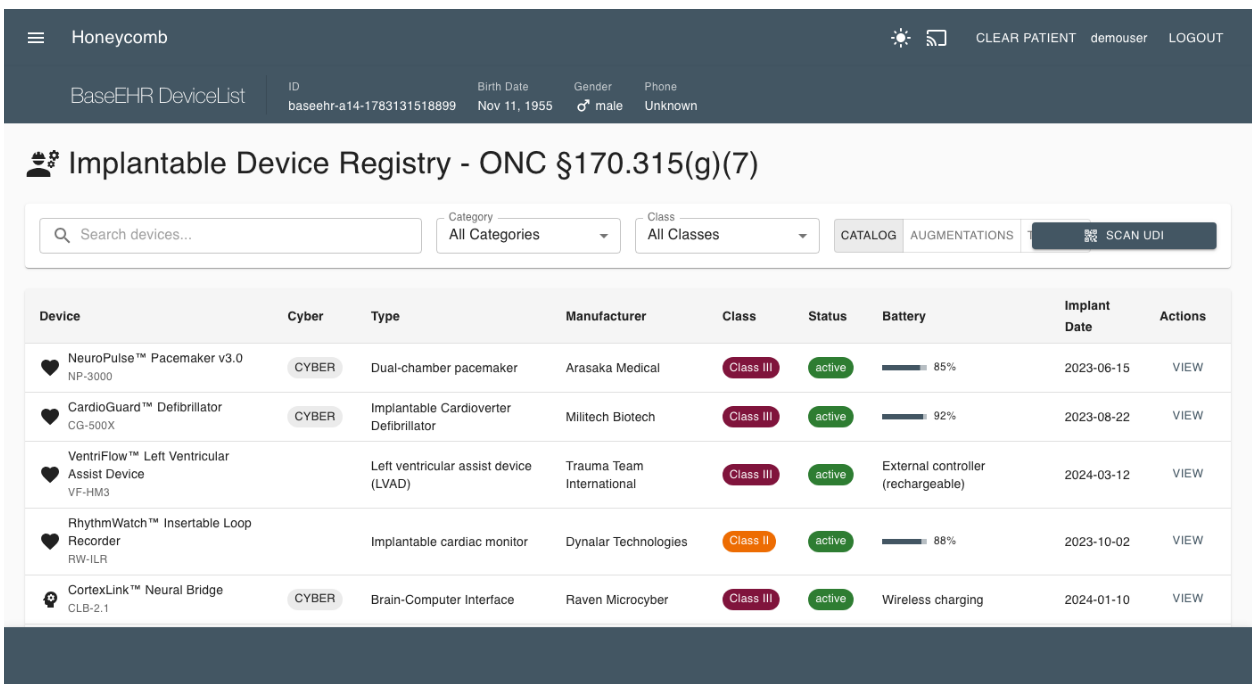

Figure 10.1: Implantable Device List - Patient View

Chapter 11

§ 170.315(e)(1) - View, Download, Transmit

11.1 Criterion Information

11.1.1 Maturity Level

IMPLEMENTED

11.1.2 Regulatory Text

From *45 CFR § 170.315(e)(1)*:

> **(1) View, download, and transmit to 3rd party.**
> The technology must enable patients to view, download, and transmit their health information.
> (Full regulatory text abbreviated. See BDD specification for complete requirements.)

11.2 Behavior-Driven Development Specification

Listing 11.1: 170.315(e)(1) - View, Download, Transmit

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-e-1-view-
    download-transmit.feature
# §170.315(e)(1) - View, Download, and Transmit to 3rd Party

# REGULATORY TEXT FROM 45 CFR §170.315(e)(1)
#
# (1) The action(s) (i.e., view, download, transmission) that occurred;

@170.315-e-1 @vdt @patient-access @patient-engagement
Feature: View, Download, and Transmit to 3rd Party
  As a patient
  I want to view, download, and transmit my health information
  So that I can access my data and share it with others

  Background:
    Given I am authenticated as a patient
    And my health information is available in the system

  Scenario: View health information online
    Given I am accessing my patient portal
    When I view my health information
    Then the system shall display my data in human readable format
    And all available USCDI data shall be viewable
    And the information shall be current and accurate

  Scenario: Download health information
    Given I am viewing my health information
    When I choose to download my data
    Then the system shall enable download
    And download shall be in machine-readable format
    And download shall include all requested data

  Scenario: Transmit to third party via encrypted email
```

```
  Given I want to share my health information
  When I transmit to a third party email address
  Then the system shall transmit encrypted
  And transmission shall be secure
  And third party shall be able to receive the data

Scenario: Transmit to third party unencrypted with patient consent
  Given I want to share my health information
  And I accept the risks of unencrypted transmission
  When I choose unencrypted transmission
  Then the system shall warn me of risks
  And system shall transmit unencrypted only with my consent
  And my choice shall be documented

Scenario: Access activity history log
  Given I have viewed, downloaded, or transmitted my data
  When I check my activity history
  Then the system shall display activity log
  And log shall show what data was accessed
  And log shall show when data was accessed
  And log shall show to whom data was transmitted
```

Chapter 12

§ 170.315(g)(7) - Application Access - Patient Selection

12.1 Criterion Information

12.1.1 Maturity Level

VERIFIED (behavioral tests green)

12.1.2 Regulatory Text

From *45 CFR § 170.315(g)(7)*:

> **(7) Application access—patient selection.**
> The technology must be able to receive a request with sufficient information to uniquely identify a patient and return an ID or other token.
> (Full regulatory text abbreviated. See BDD specification for complete requirements.)

12.2 Behavior-Driven Development Specification

Listing 12.1: 170.315(g)(7) - Application Access

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-g-7-10-apis
    .feature
# §170.315(g)(7)-(g)(10) - Application Programming Interfaces

# REGULATORY TEXT FROM 45 CFR §170.315(g)(7)
#
# (7) Application access--patient selection. The following technical outcome and
    conditions must be met through the demonstration of an application programming
    interface (API).
# (i) Functional requirement. The technology must be able to receive a request with
    sufficient information to uniquely identify a patient and return an ID or other token
     that can be used by an application to subsequently execute requests for that patient
    's data.
# (ii) Documentation.
# (A) The API must include accompanying documentation that contains, at a minimum:

@170.315-g-7-10 @api @fhir @design-performance
Feature: Application Programming Interfaces
  As a health IT developer
  I want to provide standardized APIs
  So that applications can access health information

  Background:
    Given the Health IT Module provides APIs
    And API access is required

  Scenario: Provide patient selection API (g)(7)
    Given applications need to select patients
    When application requests patient selection
    Then API shall enable patient selection
    And API shall conform to standards
    And API shall be properly documented
```

```
Scenario: Provide data category request API (g)(8)
  Given applications need specific data categories
  When application requests data by category
  Then API shall provide data category request capability
  And API shall return requested data
  And API shall conform to USCDI

Scenario: Provide all data request API (g)(9)
  Given applications need complete patient data
  When application requests all patient data
  Then API shall provide all available data
  And data shall include all USCDI elements
  And API shall conform to standards

Scenario: Provide standardized API for patient and population services (g)(10)
  Given applications need patient and population data
  When application uses FHIR API
  Then API shall conform to §170.215(a) standard
  And API shall support SMART App Launch
  And API shall support bulk data access
  And API shall provide comprehensive functionality
```

Chapter 13

§ 170.315(g)(9) - Application Access - All Data

13.1 Criterion Information

13.1.1 Maturity Level

VERIFIED (behavioral tests green)

13.1.2 Regulatory Text

From *45 CFR § 170.315(g)(9)*:

> **(9) Application access—all data request.**
> Provide an application programming interface (API) that can return all available data for a specific patient in a single request.
> (Full regulatory text abbreviated. See BDD specification for complete requirements.)

13.2 Behavior-Driven Development Specification

Listing 13.1: 170.315(g)(9) - Application Access

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-g-7-10-apis
    .feature
# §170.315(g)(7)-(g)(10) - Application Programming Interfaces

# REGULATORY TEXT FROM 45 CFR §170.315(g)(7)
#
# (7) Application access--patient selection. The following technical outcome and
    conditions must be met through the demonstration of an application programming
    interface (API).
# (i) Functional requirement. The technology must be able to receive a request with
    sufficient information to uniquely identify a patient and return an ID or other token
     that can be used by an application to subsequently execute requests for that patient
    's data.
# (ii) Documentation.
# (A) The API must include accompanying documentation that contains, at a minimum:

@170.315-g-7-10 @api @fhir @design-performance
Feature: Application Programming Interfaces
  As a health IT developer
  I want to provide standardized APIs
  So that applications can access health information

  Background:
    Given the Health IT Module provides APIs
    And API access is required

  Scenario: Provide patient selection API (g)(7)
    Given applications need to select patients
    When application requests patient selection
    Then API shall enable patient selection
    And API shall conform to standards
    And API shall be properly documented
```

```gherkin
Scenario: Provide data category request API (g)(8)
  Given applications need specific data categories
  When application requests data by category
  Then API shall provide data category request capability
  And API shall return requested data
  And API shall conform to USCDI

Scenario: Provide all data request API (g)(9)
  Given applications need complete patient data
  When application requests all patient data
  Then API shall provide all available data
  And data shall include all USCDI elements
  And API shall conform to standards

Scenario: Provide standardized API for patient and population services (g)(10)
  Given applications need patient and population data
  When application uses FHIR API
  Then API shall conform to §170.215(a) standard
  And API shall support SMART App Launch
  And API shall support bulk data access
  And API shall provide comprehensive functionality
```

Chapter 14

§ 170.315(g)(10) - Standardized API

14.1 Criterion Information

14.1.1 Maturity Level

IMPLEMENTED

14.1.2 Regulatory Text

From *45 CFR § 170.315(g)(10)*:

> **(10) Standardized API for patient and population services.**
> The technology must implement a standardized API that conforms to FHIR R4, US Core, SMART App Launch, OpenID Connect, and SMART Backend Services.
> (Full regulatory text abbreviated. See BDD specification for complete requirements.)

14.1.3 Inferno Validation Status

Conformance for this criterion is established by the ONC-approved **Inferno** testing tool, not by the in-repo behavioral suite. The Inferno (g)(10) Standardized API test suite has been run against this system with passing results; a refreshed run will be executed and its results formally recorded as part of an ACB submission. Until certification is issued by an ONC-ACB, this status represents validation evidence, not a certification claim.

14.2 Behavior-Driven Development Specification

Listing 14.1: 170.315(g)(10) - Standardized API

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-g-7-10-apis
    .feature
# §170.315(g)(7)-(g)(10) - Application Programming Interfaces

# REGULATORY TEXT FROM 45 CFR §170.315(g)(7)
#
# (7) Application access--patient selection. The following technical outcome and
    conditions must be met through the demonstration of an application programming
    interface (API).
# (i) Functional requirement. The technology must be able to receive a request with
    sufficient information to uniquely identify a patient and return an ID or other token
     that can be used by an application to subsequently execute requests for that patient
    's data.
# (ii) Documentation.
# (A) The API must include accompanying documentation that contains, at a minimum:

@170.315-g-7-10 @api @fhir @design-performance
Feature: Application Programming Interfaces
  As a health IT developer
  I want to provide standardized APIs
  So that applications can access health information

  Background:
    Given the Health IT Module provides APIs
```

```
    And API access is required

  Scenario: Provide patient selection API (g)(7)
    Given applications need to select patients
    When application requests patient selection
    Then API shall enable patient selection
    And API shall conform to standards
    And API shall be properly documented

  Scenario: Provide data category request API (g)(8)
    Given applications need specific data categories
    When application requests data by category
    Then API shall provide data category request capability
    And API shall return requested data
    And API shall conform to USCDI

  Scenario: Provide all data request API (g)(9)
    Given applications need complete patient data
    When application requests all patient data
    Then API shall provide all available data
    And data shall include all USCDI elements
    And API shall conform to standards

  Scenario: Provide standardized API for patient and population services (g)(10)
    Given applications need patient and population data
    When application uses FHIR API
    Then API shall conform to §170.215(a) standard
    And API shall support SMART App Launch
    And API shall support bulk data access
    And API shall provide comprehensive functionality
```

Chapter 15

§ 170.315(c)(1) - Clinical Quality Measures - Record and Export

15.1 Criterion Information

15.1.1 Maturity Level

VERIFIED (behavioral tests green)

15.1.2 Regulatory Text

From *45 CFR § 170.315(c)(1)*:

> **(1) Clinical quality measures — record and export.**
> **(i) Record.** For each clinical quality measure (CQM) presented for certification, record all of the data identified in the standard necessary to calculate the measure.
> **(ii) Export.** Enable a user to electronically export a data file, for one or more patients, formatted per the QRDA Category I standard (45 CFR 170.205(h)(2)), that includes all of the recorded CQM data.

(Full regulatory text abbreviated. See BDD specification for complete requirements.)

15.1.3 Implementation

The behavioral test verifies the record-calculate-export pipeline: the measures catalog is populated (seeded CMS/PACIO measures); a numerator event is recorded as a measure-observation-profiled FHIR `Observation`; an individual FHIR `MeasureReport` is calculated (fqm-execution / PACIO evaluators) with initial-population, denominator, and numerator groups and persisted; and the captured reports export as a FHIR Bundle, CSV, JSON, **and QRDA Category I** (§ 170.205(h)(2)). The QRDA I export produces a patient-level CDA document with the QRDA Category I template identifiers and Measure, Reporting Parameters, and Patient Data sections, built from the patient's real demographics and the measure's population results.

Scope: the QRDA I document is real and standards-shaped (correct envelope, template identifiers, and sections). Full per-measure QDM data-criteria entry coverage validated by the Cypress tool is the remaining external certification work; QRDA Category III (for (c)(3)) is a follow-on.

15.2 Behavior-Driven Development Specification

Listing 15.1: 170.315(c)(1) - CQM Record and Export

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-c-1-cqm-
    record-export.feature
# §170.315(c)(1) - Clinical Quality Measures - Record and Export

# REGULATORY TEXT FROM 45 CFR §170.315(c)(1)
#
# (1) Clinical quality measures--record and export --
# (i) Record. For each and every CQM for which the technology is presented for
    certification, the technology must be able to record all of the data that would be
    necessary to calculate each CQM. Data required for CQM exclusions or exceptions must
    be codified entries, which may include specific terms as defined by each CQM, or may
    include codified expressions of "patient reason," "system reason," or "medical reason
    ."
```

```gherkin
# (ii) Export. A user must be able to export a data file at any time the user chooses and
    without subsequent developer assistance to operate:
# (A) Formatted in accordance with the standard specified in §170.205(h)(2);
# (B) Ranging from one to multiple patients; and
# (C) That includes all of the data captured for each and every CQM to which technology
    was certified under paragraph (c)(1)(i) of this section.

@170.315-c-1 @cqm-record-export @clinical-quality-measures @cqm
Feature: Clinical Quality Measures - Record and Export
  As a healthcare provider
  I want to record and export clinical quality measure data
  So that I can track quality metrics and submit quality reports

  Background:
    Given I am authenticated as a provider
    And the application has the clinics:quality-measures package installed

  Scenario: Record all data necessary for CQM calculation
    Given I am documenting patient care for a certified CQM
    When I record clinical data elements
    Then the system shall enable recording of all data necessary to calculate each CQM
    And all required data elements for the CQM shall be capturable
    And the data shall be stored in structured format

  Scenario: Record codified CQM exclusion data
    Given I am documenting a CQM exclusion
    When I enter the exclusion reason
    Then the system shall accept codified entries for exclusions
    And codified entries may include specific terms defined by the CQM
    And codified entries may include "patient␣reason" expression
    And codified entries may include "system␣reason" expression
    And codified entries may include "medical␣reason" expression

  Scenario: Record codified CQM exception data
    Given I am documenting a CQM exception
    When I enter the exception reason
    Then the system shall accept codified entries for exceptions
    And codified entries may include specific terms defined by the CQM
    And codified entries may include reason expressions per CQM requirements

  Scenario: Export CQM data at user's discretion
    Given CQM data has been recorded for patients
    When I choose to export CQM data
    Then the system shall enable export at any time I choose
    And export shall not require subsequent developer assistance
    And I shall be able to initiate export independently

  Scenario: Export CQM data in QRPP format
    Given CQM data has been recorded
    When I export the data
    Then the export shall be formatted per §170.205(h)(2)
    And the format shall be Quality Reporting Document Architecture QRPP

  Scenario: Export CQM data for single or multiple patients
```

```
  Given CQM data exists for multiple patients
  When I configure the export
  Then the system shall enable export ranging from one to multiple patients
  And I shall be able to select specific patients
  Or I shall be able to export data for patient populations

Scenario: Export includes all captured CQM data
  Given data has been captured for certified CQMs
  When I export CQM data
  Then the export shall include all data captured for each certified CQM
  And no required data elements shall be omitted from export
  And the export shall be complete and accurate
```

15.3 Screenshots

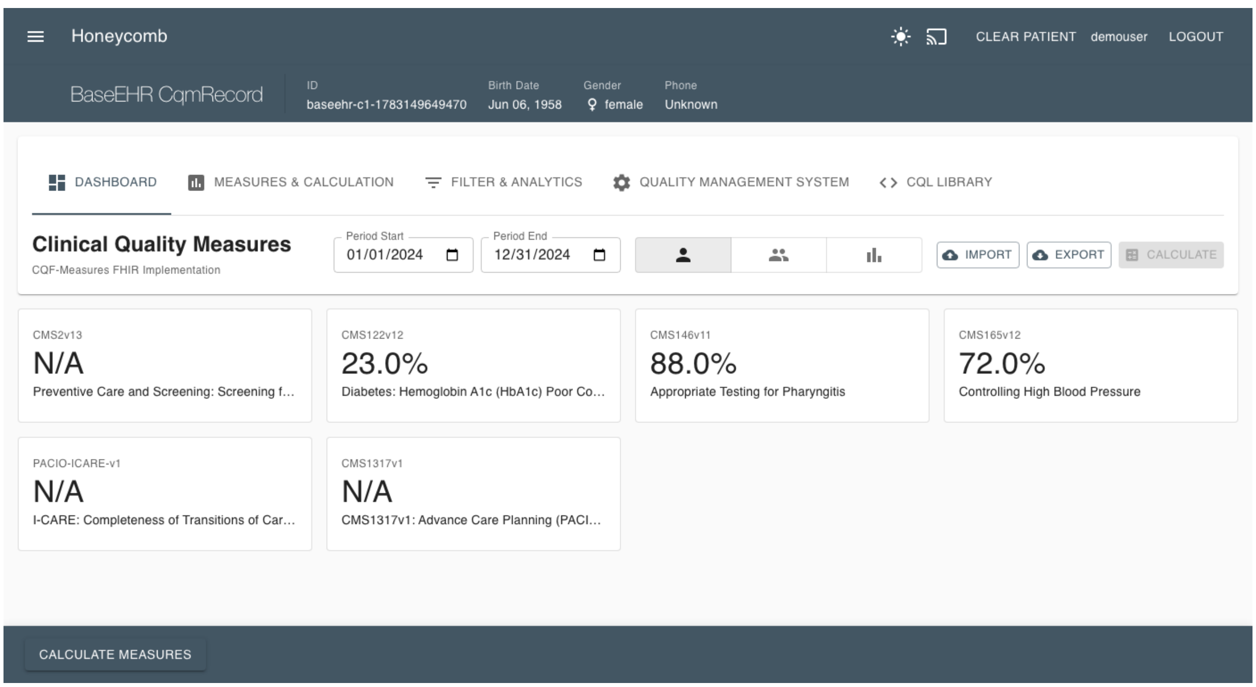

Figure 15.1: Clinical Quality Measures - Record and Export

Chapter 16

§ 170.315(h)(1) - Direct Project

16.1 Criterion Information

16.1.1 Maturity Level

GAP (documented, behavioral tests red)

16.1.2 Regulatory Text

From *45 CFR § 170.315(h)(1)*:

> **(1) Direct Project.**
> Send and receive health information using the Direct Standard (Applicability Statement for Secure Health Transport), including SMTP with S/MIME message signing and encryption, message disposition notifications, and trust anchored in X.509 certificates. Criterion (h)(2) additionally requires the Edge Protocols (XDR/XDM) alongside Direct.

(Full regulatory text abbreviated. See BDD specification for complete requirements.)

16.1.3 Documented Gap

The behavioral test verifies the messaging scaffold: the Direct messaging routes render; a Direct address is format-validated (and a malformed address is rejected); and a sent Direct message is persisted as a FHIR `Communication` with the Direct-Message category and an encryption extension, retrievable afterward. These pass green.

Two capabilities are **gaps**, and this is the expected headline Base-EHR gap: (1) there is **no operational Direct transport** — no SMTP+S/MIME send/receive through a Health Information Service Provider (HISP); the configured relay, when present, is plain SMTP, not the Direct Applicability Statement stack (S/MIME signing/encryption, message disposition notifications, DNS CERT / LDAP certificate discovery), and messages persist only as FHIR resources; and (2) **trust verification is simulated** — a placeholder "Demo Certificate Authority" certificate is returned for any well-formed address, with no real X.509 chain validation or DirectTrust bundle. The Edge Protocols (XDR/XDM) required by (h)(2) have no implementation.

Remediation: integrate a real Direct/HISP stack (SMTP + S/MIME per the Applicability Statement, MDN handling, trust-bundle-based certificate validation), and add XDR/XDM for (h)(2). Certification testing uses the ONC Direct Transmission Testing (DTT) tool against a live HISP.

16.2 Behavior-Driven Development Specification

Listing 16.1: 170.315(h)(1) - Direct Project

```
# certification/bdd/170.315-h-1-direct-project.feature
# REGULATORY TEXT FROM 45 CFR §170.315(h)(1)
#
# (1) Direct Project --
# (i) Applicability Statement for Secure Health Transport. Able to send and receive
    health information in accordance with the standard specified in §170.202(a)(2),
    including formatted only as a "wrapped" message.
```

```
# (ii) Delivery Notification in Direct. Able to send and receive health information in
    accordance with the standard specified in §170.202(e)(1).

@170.315-h-1 @direct @transport
Feature: Direct Project
  As a healthcare provider
  I want to send and receive health information via Direct messaging
  So that I can securely exchange patient information with other providers

  Background:
    Given the system supports Direct Project messaging
    And I have appropriate Direct addressing

  Scenario: Send health information via Direct with wrapped message
    Given I have health information to send
    When I send information via Direct messaging
    Then the system shall send per §170.202(a)(2)
    And the message shall be formatted as a wrapped message

  Scenario: Receive health information via Direct with wrapped message
    Given health information is being sent to me
    When I receive the Direct message
    Then the system shall receive per §170.202(a)(2)
    And the system shall process the wrapped message

  Scenario: Send delivery notification in Direct
    Given I have sent health information via Direct
    When delivery notification is enabled
    Then the system shall send delivery notification per §170.202(e)(1)

  Scenario: Receive delivery notification in Direct
    Given I sent health information via Direct
    When a delivery notification is sent back
    Then the system shall receive delivery notification per §170.202(e)(1)
```

16.3 Screenshots

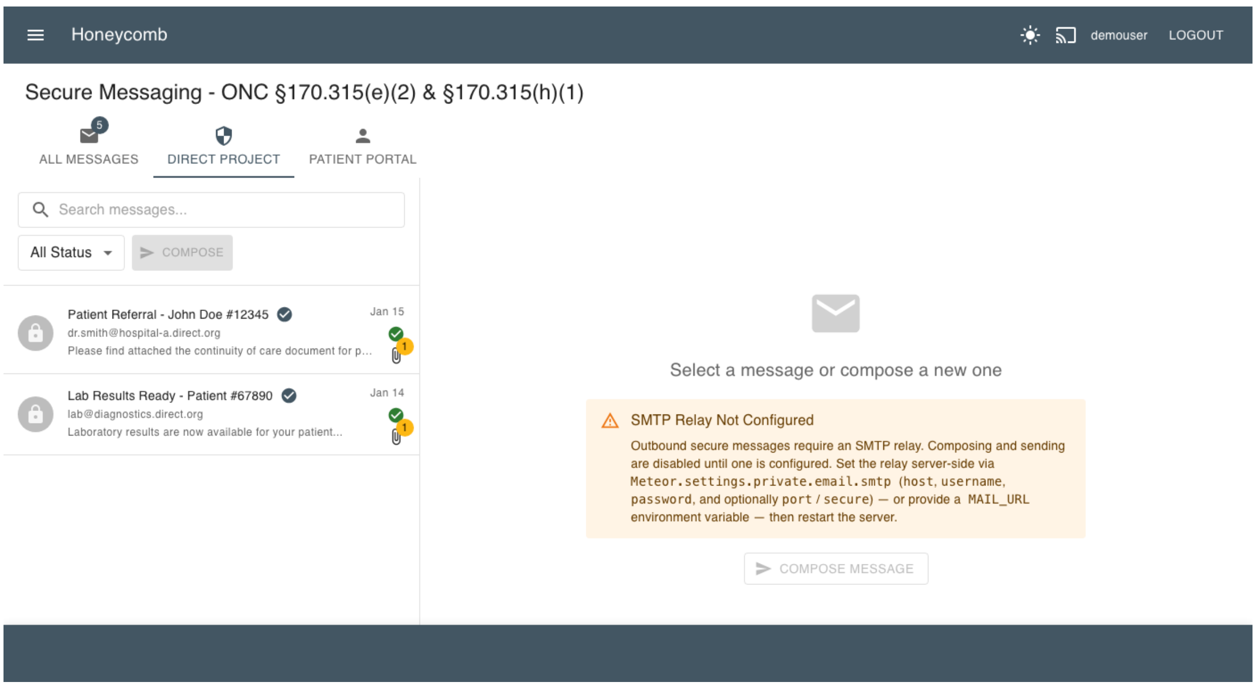

Figure 16.1: Direct Project - Secure Messaging

Appendix A

Acronyms and Abbreviations

API
: Application Programming Interface

BDD
: Behavior-Driven Development

C-CDA
: Consolidated Clinical Document Architecture

CDS
: Clinical Decision Support

CFR
: Code of Federal Regulations

CPOE
: Computerized Provider Order Entry

DI Device Identifier

DSI Decision Support Interventions

EHR
: Electronic Health Record

FHIR
: Fast Healthcare Interoperability Resources

GMDN
: Global Medical Device Nomenclature

GUDID
: Global Unique Device Identification Database

HCT/P
: Human Cells, Tissues, and Cellular and Tissue-Based Products

HL7
: Health Level Seven

MRI
Magnetic Resonance Imaging

OAuth
Open Authorization

ONC
Office of the National Coordinator for Health Information Technology

PI Production Identifier

SMART
Substitutable Medical Applications, Reusable Technologies

SNOMED CT
Systematized Nomenclature of Medicine - Clinical Terms

SPCU
Sex Parameter for Clinical Use

UDI
Unique Device Identifier

USCDI
United States Core Data for Interoperability

VDT
View, Download, Transmit

Appendix B

References

1. Office of the National Coordinator for Health Information Technology. *2015 Edition Health IT Certification Criteria, 2015 Edition Base EHR Definition, and ONC Health IT Certification Program Modifications*. 45 CFR Part 170. Federal Register, 2015.

2. Office of the National Coordinator for Health Information Technology. *21st Century Cures Act: Interoperability, Information Blocking, and the ONC Health IT Certification Program*. 45 CFR Part 170. Federal Register, 2020.

3. HL7 International. *FHIR R4 Specification*. http://hl7.org/fhir/R4/

4. HL7 International. *US Core Implementation Guide*. http://hl7.org/fhir/us/core/

5. SMART Health IT. *SMART App Launch Framework*. http://hl7.org/fhir/smart-app-launch/

6. Office of the National Coordinator for Health Information Technology. *United States Core Data for Interoperability (USCDI) Version 5*. 2024.

7. FDA. *Unique Device Identification System (UDI System)*. 21 CFR Part 801.

8. FDA. *Global Unique Device Identification Database (GUDID)*. https://accessgudid.nlm.nih.gov/

Appendix C

Certification Testing Notes

C.1 Behavioral Test Suite Status

Each Base EHR criterion in Part I has a behavioral Nightwatch test under `certification/tdd/base_ehr/` (one file per criterion, 170.315.<x>.<y>.test.js). The suite drives real record/change/access behaviors against FHIR resources per the ONC test procedures, not mere page presence, and is run against a live server. As of this revision the outcome is **11 verified** (green) and **1 documented gap** (red): (a)(1), (a)(2), (a)(3), (a)(5), (a)(14), (b)(1), (b)(4), (b)(11), (c)(1), (g)(7), and (g)(9) verify; only (h)(1) Direct Project carries a documented gap. Any remaining gaps use explicit `GAP(170.315.x.y)` annotations that fail deliberately, so the suite functions as a live certification punch-list; the Status column in the Overview and the per-criterion chapters reflect these results. Criterion (g)(10) is covered externally by Inferno (below), not by this suite.

C.2 Inferno Testing

For comprehensive certification testing of the standardized API criterion (g)(10) — and to authoritatively validate the API surfaces exercised by (g)(7) and (g)(9) — the official ONC-approved Inferno testing tool must be used. The Inferno (g)(10) suite has been run against this system with passing results; a refreshed run will be executed and formally recorded as part of an ACB submission.

Inferno Website: https://inferno.healthit.gov

Inferno provides authoritative test suites that validate:

- FHIR R4 conformance
- US Core Implementation Guide compliance
- SMART App Launch OAuth 2.0 flows
- OpenID Connect Core
- SMART Backend Services (Bulk FHIR)
- USCDI data completeness
- Token management and revocation
- All mandatory and must-support elements

The Nightwatch tests in this system provide basic endpoint verification for development and CI/CD purposes. They are **not** a replacement for Inferno certification testing.

C.3 Test Execution

To run the complete Base EHR test suite:

```
cd /path/to/core
chmod +x certification/tdd/base_ehr/run-base-ehr-tests.sh
./certification/tdd/base_ehr/run-base-ehr-tests.sh
```

Or run a single criterion directly against a running server:

```
npx nightwatch --config nightwatch.circle.conf.js
  certification/tdd/base_ehr/170.315.<x>.<y>.test.js
```

The suite covers the twelve in-scope CY2026 Base EHR criteria (one test file each); (g)(10) is intentionally excluded and left to Inferno.

Appendix D

Software Bill of Materials (SBOM)

This appendix is the machine-derived Software Bill of Materials for Care Commons EHR. It is generated from the committed `package-lock.json` (`lockfileVersion 3`) — the exact dependency graph resolved for the certification-candidate build — so it documents the software as *actually assembled*, not as aspirationally specified. A complete supply-chain manifest is a deliberate design output: it lets a reviewer, integrator, or downstream deployer reason about provenance, patch exposure, and license obligation without treating the system as a black box.

D.1 SBOM Metadata

Field	Value
Document format	Dependency manifest (npm `lockfileVersion 3`)
Enumerated on	2026-07-04
Source of truth	`package-lock.json` at the certification commit
Certification commit	see Reproducibility (§0.7)
Build runtime (Node)	22.22.0 (Meteor bundled)
Lockfile tooling (npm)	11.12.1
Framework	Meteor 3.4
Total resolved packages	4,121
Direct dependencies (runtime)	108
Direct dependencies (dev)	12
First-party workspace packages	70
Third-party with SPDX metadata	4,003
Packages without SPDX metadata	118

Methodology note. Versions below are reported *exactly as resolved* in the lockfile. Because Care Commons pins several transitive dependencies through explicit `overrides` (for security back-ports) and runs on the bundled Meteor toolchain, a resolved version may differ from a package's latest public-registry release. This is expected: the SBOM records the shipped artifact, which is the artifact put forward for certification.

D.2 Key Components

The following are the load-bearing third-party components — the packages a reviewer most needs to understand. The full transitive set (4,121 entries) lives in `package-lock.json`; its integrity hash is recorded in the Reproducibility table (§0.7).

Package	**Version**	**Purpose**	**License**
`react` / `react-dom`	18.3.1	UI rendering	MIT
`@mui/material`	5.18.0	Component / design system	MIT
`@mui/icons-material`	5.18.0	Icon set	MIT
`mongodb`	6.21.0	FHIR resource persistence	Apache-2.0
`express`	4.22.1	HTTP / REST routing (FHIR server)	MIT
`@node-oauth/express-oauth-server`	4.1.5	OAuth2 authorization server (SMART)	MIT
`jsonwebtoken`	9.0.3	JWT signing / verification	MIT
`jose`	6.2.3	JWK / JWS / JWE (Backend Services)	MIT
`pkijs`	3.3.3	X.509 / PKI (Direct, key material)	BSD-3-Clause
`asn1js`	3.0.7	ASN.1 encoding (certificates)	BSD-3-Clause
`dcmjs`	0.38.3	DICOM parse / serialize	MIT
`@cornerstonejs/core`	1.86.1	Diagnostic image viewing	MIT
`@cornerstonejs/dicom-image-loader`	1.86.0	DICOM image loader	MIT
`fqm-execution`	1.8.5	FHIR quality-measure engine (c.1)	Apache-2.0
`xml2js`	0.6.2	C-CDA / QRDA XML parse (b.1, c.1)	MIT
`fhirpath`	4.8.2	FHIRPath evaluation (search, validation)	Apache-2.0
`simpl-schema`	3.4.6	FHIR resource schema validation	MIT
`lodash`	4.18.1	Defensive get/set circuit breaker	MIT
`moment`	2.30.1	Date / time handling	MIT
`nightwatch`	3.14.0	Behavioral certification E2E tests	MIT
`chromedriver`	146.0.6	Headless browser driver (E2E)	Apache-2.0
`@rspack/core`	1.7.1	Workflow-package bundler	MIT

D.3 License Distribution (raw)

The following is the verbatim SPDX-identifier distribution across all 4,003 third-party packages that declare license metadata, as scanned from the lockfile. It is presented un-normalized for full transparency; the normalized compliance analysis is in Appendix E.

Count	SPDX identifier
2919	MIT
283	Apache-2.0 OR MIT
273	ISC
171	BSD-3-Clause
129	Apache-2.0
70	UNLICENSED (first-party workspace)
40	0BSD
35	BSD-2-Clause
23	BlueOak-1.0.0
13	MPL-2.0
5	(MIT OR CC0-1.0)
5	(Apache-2.0 OR MIT)
4	Unlicense
4	(MIT AND Zlib)
3	(MIT OR Apache-2.0)
3	(WTFPL OR MIT)
2	SEE LICENSE IN LICENSE.md
2	SEE LICENSE IN LICENSE
2	WTFPL OR ISC
2	(MIT AND BSD-3-Clause)
1	Artistic-2.0
1	MIT-0
1	(Unlicense OR Apache-2.0)
1	Python-2.0
1	CC-BY-4.0
1	(MIT OR WTFPL)
1	CC0-1.0
1	(MIT OR GPL-3.0-or-later)
1	LGPL-3.0
1	(BSD-3-Clause OR GPL-2.0)
1	(BSD-2-Clause OR MIT OR Apache-2.0)
1	WTFPL

Appendix E

License Compliance Report

This appendix normalizes the raw distribution in Appendix D into license *families*, states the obligation each family imposes, and records how those obligations interact with Care Commons EHR's own distribution license. The headline finding: the dependency tree is overwhelmingly permissive (roughly 96% MIT / ISC / BSD / Apache / public-domain-equivalent), carries no imported strong-copyleft (GPL/AGPL) obligation, and every copyleft component present is either file-level (MPL-2.0) or weak (LGPL-3.0) and used in a compatible way.

E.1 The Application's Own License

The Care Commons EHR application code is licensed **AGPL-3.0** (see `LICENSE.md`). This is an intentional network-copyleft posture: an operator running a modified Care Commons EHR as a network service must offer that modified source to its users. AGPL-3.0 is the *outbound* license of the first-party application; it is not imposed by any dependency. The workflow packages under **`npmPackages/`** are individually MIT (or Apache-2.0 for the core subset); private packages under **`extensions/`** are `UNLICENSED` and are not distributed with the open-source release.

E.2 License Families and Obligations

Family	**Count**	**Obligation**	**Compat.**
MIT / ISC / 0BSD / BSD-2 / BSD-3 / BlueOak	3462	Preserve copyright + license text (attribution)	✓
Apache-2.0 (incl. Apache/MIT dual)	420	Preserve NOTICE; explicit patent grant	✓
Public-domain-equivalent (Unlicense, CC0, MIT-0)	12	None	✓
Other permissive dual/combos (Zlib, WTFPL, Python-2.0, . . .)	16	Attribution where applicable	✓
MPL-2.0	14	File-level source disclosure for *modified* MPL files only	✓
LGPL-3.0	1	Allow relinking; provide LGPL source (dynamic use)	✓

Family	Count	Obligation	Compat.
Dual-with-GPL (resolve to permissive side)	2	Elect the non-GPL option (MIT / BSD)	✓
CC-BY-4.0	1	Attribution (asset / data, not code)	✓
Artistic-2.0	2	Attribution; state changes	✓
Custom ("SEE LICENSE IN...")	5	Manual review of bundled LICENSE file	review
UNLICENSED (first-party workspace)	70	N/A — own code (AGPL / MIT / private)	—
No SPDX metadata	118	Mostly `@types` stubs + first-party; reviewed	low

E.3 Attribution and Notice Obligations

- **Permissive attribution (MIT/BSD/ISC).** The largest obligation by count is simple attribution: the copyright notice and license text of each package must travel with any redistribution. These are preserved in each package's own `LICENSE` file within `node_modules` and are reproduced in binary/desktop distributions.
- **Apache-2.0 NOTICE.** Apache-2.0 packages that ship a `NOTICE` file require that file to be preserved and passed through; Apache-2.0 also grants an explicit patent license, which is favorable for a regulated deployment.
- **MPL-2.0 (file-level copyleft).** MPL is per-file: only if a specific MPL-licensed *source file* is modified must that file's source be offered. Care Commons consumes these packages unmodified, so the only obligation is to preserve their license headers. MPL-2.0 §3.3 is explicitly compatible with (A)GPL distribution.
- **LGPL-3.0 (weak copyleft).** The single LGPL component is used as a dynamically-resolved library, satisfying the relinking allowance; its source is available upstream.
- **Dual-licensed with a GPL option.** Two packages offer a choice that includes a GPL variant; Care Commons elects the permissive alternative (MIT / BSD-3-Clause), so no GPL obligation attaches.

E.4 Source Availability

Because the application is AGPL-3.0, corresponding source for the first-party code is made available with every network deployment. Third-party source is obtainable from each package's public repository (recorded in `package-lock.json` `resolved` URLs); nothing in the tree is a proprietary binary blob requiring a separate source-offer letter.

E.5 Findings

1. **No imported strong copyleft.** There is no GPL-2.0-only or GPL-3.0-only or AGPL *dependency* in the tree. The only AGPL in the system is Care Commons EHR's own outbound license.

2. **No external UNLICENSED risk.** All 70 `UNLICENSED` entries are first-party workspace packages (verified: every one resolves to `npmPackages/`, `extensions/`, or `core/` via a workspace symlink). None is an unlicensed third-party import.

3. **Copyleft is bounded.** The 14 MPL-2.0 and 1 LGPL-3.0 components impose only file-level / relinking obligations, all satisfied by unmodified dynamic use.

4. **Flagged for periodic manual review.** 5 "SEE LICENSE" custom declarations and 118 metadata-absent packages (predominantly `@types` stubs, which are non-executable) should be re-examined on each major dependency bump. None is currently believed to impose a restrictive obligation.

E.6 Open-Source Provenance

Beyond "which licenses," the more useful transparency question is *why each major component exists*. The table below turns the dependency list into an architectural map, so a reviewer can see the system as an intentional assembly of well-understood parts.

Component	**Reason it is in the system**
Meteor 3.4 (runtime)	Full-stack framework: DDP transport, reactive data, build toolchain
React + React-DOM	Client UI rendering
Material-UI (MUI)	Component library and light/dark design system
MongoDB driver	Document persistence for FHIR resources
Express	HTTP surface for the FHIR REST API
@node-oauth/express-oauth-server	OAuth2 authorization server for SMART App Launch
jsonwebtoken + jose	JWT / JWK handling for SMART and Backend Services
pkijs + asn1js	X.509 / certificate handling for the Direct transport and key material
dcmjs	DICOM parsing and serialization
Cornerstone3D	Diagnostic image viewing
fqm-execution	FHIR Clinical Quality Measure calculation (§170.315(c)(1))
xml2js	C-CDA and QRDA XML parsing (§170.315(b)(1), (c)(1))
fhirpath	FHIRPath evaluation for search and validation
simpl-schema	FHIR resource schema validation
Nightwatch	Behavioral compliance (E2E) testing
Inferno (external)	ONC-approved standardized-API conformance testing (§170.315(g)(10))

E.7 Reproducible Build

So that another engineer can recreate the exact tree audited above, the canonical build coordinates are recorded once in the Reproducibility table (§0.7) and summarized here: Git commit SHA, `package-lock.json` SHA-256 integrity hash, Node 22.22.0, npm 11.12.1, Meteor 3.4, MongoDB 7.0.16 (dev) / 6.0 (CI), on macOS / Linux. Restoring that lockfile with `npm ci` yields a byte-identical dependency graph, and therefore an SBOM identical to this one.